

CITY OF CHICAGO TAXI TRIPS

SHRUTHI SURESH



PROJECT DESCRIPTION AND SCOPE

Analyze Taxi Trip Trends & Performance

- Identify patterns in trip duration, fare distribution, and peak hours across Chicago.
- Understand the impact of trip cancellations, payment types, and passenger count on revenue.

Optimize Taxi Operations & Revenue

- Provide insights to improve dispatching efficiency and reduce idle time.
- Help taxi companies optimize pricing strategies based on time, location, and demand.

Enhance City Transportation & Rider Experience

- Assist policymakers in understanding taxi demand trends for better city planning.
- Improve passenger experience by analyzing fare efficiency and trip availability.

FINAL GOAL OF THE PROJECT

- **Understand taxi trip behavior** by analyzing patterns in ride frequency, duration, and fare variability across different time periods.
- **Assess demand fluctuations** to identify peak and off-peak hours, helping optimize taxi availability and pricing strategies.
- **Enhance operational efficiency** by evaluating factors affecting trip cancellations, payment methods, and driver performance.
- **Provide actionable insights** for taxi companies and city planners to improve urban mobility and reduce inefficiencies.
- **Leverage BI & Data Warehousing techniques** to create an interactive dashboard that supports informed decision-making for fleet management and service improvements.

SOFTWARE USED



Data Processing & Cleaning

- Python – Used for data transformation, cleaning, and handling missing values.
- Pandas – Data manipulation and preprocessing.
- Jupyter Notebook – Interactive environment for exploratory data analysis

Data Storage & Querying

- MySQL – Used for storing structured taxi trip data and executing SQL queries.

ETL & Data Transformation

- SQL Queries – Implemented to extract, transform, and load (ETL) data.
- Pandas & NumPy – Applied for data transformations and aggregations.

Data Visualization & Analysis

- Tableau – Developed dashboards for interactive BI analysis and visualization.

Data Modeling & Star Schema Design

- Lucidchart – Created schema diagrams for fact and dimension tables.

DATA & DATA SOURCES

Data Overview

- Size: 6.4+ million trip records
- Timeframe: Full-year 2024 dataset
- Format: CSV, structured for analysis

Key Features

- Trip Details: Start & end timestamps, trip duration, total fare
- Location Data: Pickup & drop-off community areas, GPS coordinates
- Payment Insights: Cash, card, mobile payments, tolls, and surcharges
- Operational Metrics: Trip distances, passenger count, taxi company

Data Collection

- Source: Taxi meters integrated with GPS & payment systems
- Sampling: Census-style collection of all trips in 2024
- Timeframe: Includes morning, afternoon, and night rides
- Demographics: Focuses on Chicago community areas & census tracts

TARGET DW DATABASE

Database Details

- DBMS: MySQL
- Schema Type: Star Schema (Fact & Dimension Tables)
- Storage Format: Relational Database

Key Components

- Fact Table (taxi_trips_fact) – Stores trip-level details (fares, duration, distance)
- Dimension Tables – Provides categorical context (time, location, payment, company)
- Optimized for BI & OLAP Queries – Enables fast data retrieval for reporting & analytics

Why MySQL?

- Scalable – Handles large datasets efficiently
- Optimized for Queries – Supports real-time analysis & aggregation
- Seamless Integration – Connects with Tableau for visualization

ETL- EXTRACTION

- **Data Loading:** Imported the Chicago Taxi Trips dataset and examined its structure.
- **Standardization:** Converted Trip Start Timestamp to datetime format and formatted Taxi ID as a string.
- **Missing Values:** Identified and handled gaps in Trip Miles, Fare Amount, and Timestamps.
- **Outlier Removal:** Filtered Trip Miles above the 99th percentile and removed unrealistic Fare Amounts.
- **Data Structuring:** Ensured correct data types for numerical and categorical fields, preparing it for further analysis.

ETL- TRANSFORMATIONS

- **Feature Engineering:** Extracted Trip Start Hour from timestamps for time-based analysis
- **Geocoding:** Converted pickup and drop off locations into latitude & longitude coordinates for spatial analysis.
- **Categorization:** Classified trips based on payment type, trip distance, and fare amount for segmentation.
- **Data Aggregation:** Grouped trips by hour, day, and company to analyze trends.
- **Encoding:** Converted categorical variables like payment type and company name into numerical values for modeling.
- **Final Structuring:** Prepared cleaned and transformed data for efficient querying, visualization, and further analysis.

ETL- LOADING

Transformed data is loaded into MySQL for structured storage.

- **Fact Table:** Stores trip-level metrics (trip duration, fare, distance, payment type).
- **Dimension Tables:** Hold metadata for time, location, and taxi companies.
- Ensures optimized querying for BI dashboards.

How it works:

- New trip records are appended to the Fact Table (taxi_trips_fact) daily.
- Dimension Tables (time_dim, location_dim, company_dim) are updated only when new values appear.
- Primary keys ensure no duplicate entries in the Fact Table.
- Indexes on timestamps allow faster queries for incremental updates.

BUSINESS AND USER EXPERIENCE

End Users of Data Warehouse (DW):

- Executives & Decision Makers – Strategic planning based on revenue trends.
- Marketing Analysts – Optimize trip-based promotions and customer segmentation.
- Sales Teams – Identify high-revenue locations and target key customers.
- Operations Managers – Improve fleet allocation and driver efficiency.
- Finance Departments – Track earnings, fare trends, and optimize pricing models.

Business Intelligence Benefits:

- Enhanced Decision Making – Use data insights for better strategic planning.
- Operational Efficiency – Optimize routes, pricing, and driver allocation.
- Customer Experience Improvement – Reduce wait times, suggest better drop-offs.
- Competitive Advantage – Utilize data-driven strategies to stay ahead.
- Risk Management – Detect fraudulent trips and ensure pricing fairness.

INCREMENTAL UPDATE OF THE STAR SCHEMA

Incremental updates to the Star Schema involve:

- Identifying new records that are not already in the taxi_trips_fact table.
- Updating dimension tables (time_dim, location_dim) if new records introduce previously unseen timestamps or locations.
- Inserting only new trips into the fact table to avoid duplicates.

Steps to Implement Incremental Updates

- Find the latest trip already inserted into taxi_trips_fact

```
SELECT MAX(t.Trip_Date) FROM taxi_trips_fact f
JOIN time_dim t ON f.Time_ID = t.Time_ID;
```

MAX(t.Trip_Date)
2025-01-01

- Insert new dates into time_dim if they don't exist

```
INSERT INTO time_dim (Trip_Date, Trip_Hour, Day_Of_Week, Month, Year)
SELECT DISTINCT DATE(Trip_Start_Timestamp), HOUR(Trip_Start_Timestamp),
    DAYNAME(Trip_Start_Timestamp), MONTH(Trip_Start_Timestamp), YEAR(Trip_Start_Timestamp)
FROM taxi_trips
WHERE Trip_Start_Timestamp > (SELECT MAX(t.Trip_Date) FROM taxi_trips_fact f
    JOIN time_dim t ON f.Time_ID = t.Time_ID);
```


INCREMENTAL UPDATE OF THE STAR SCHEMA

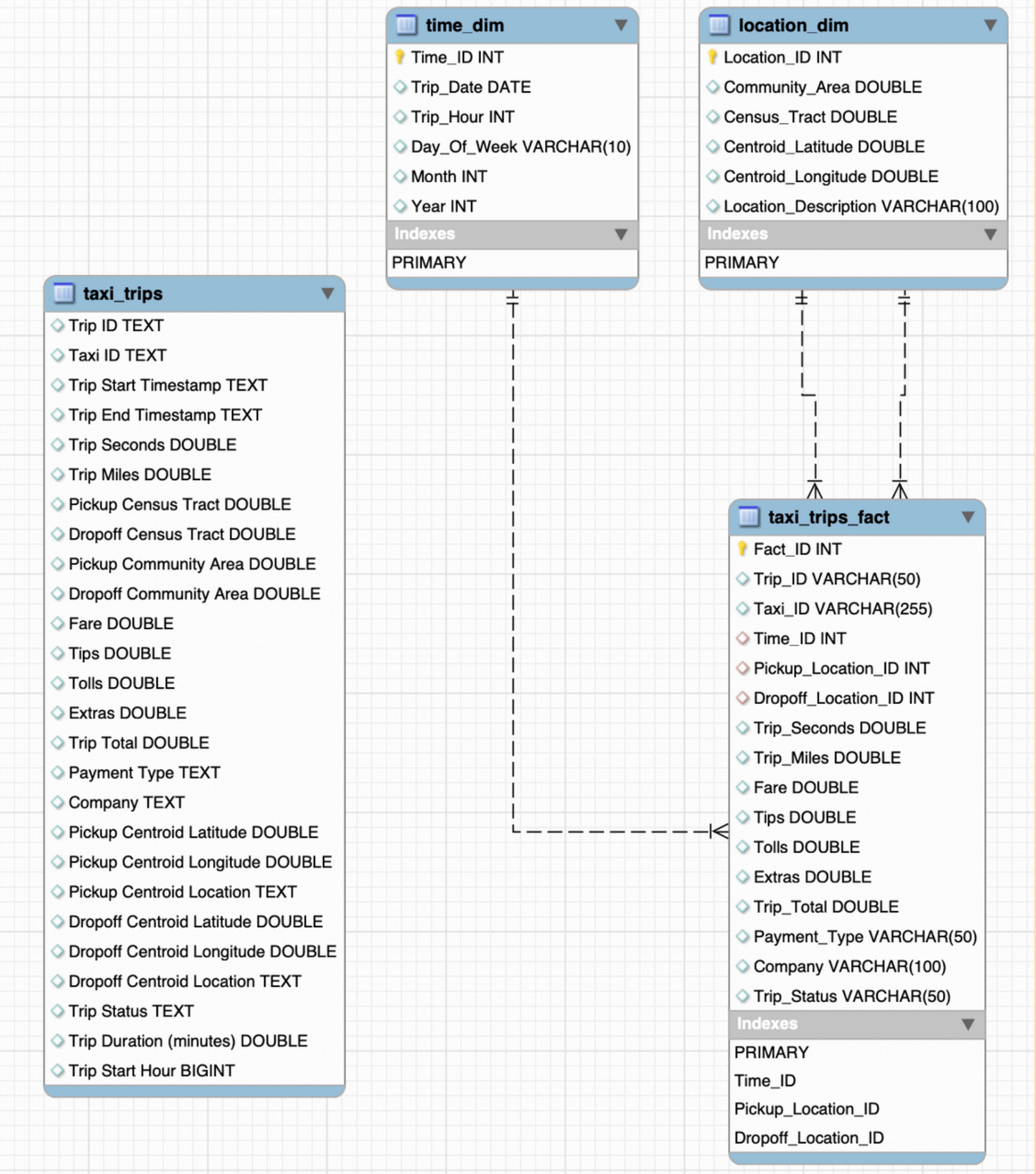
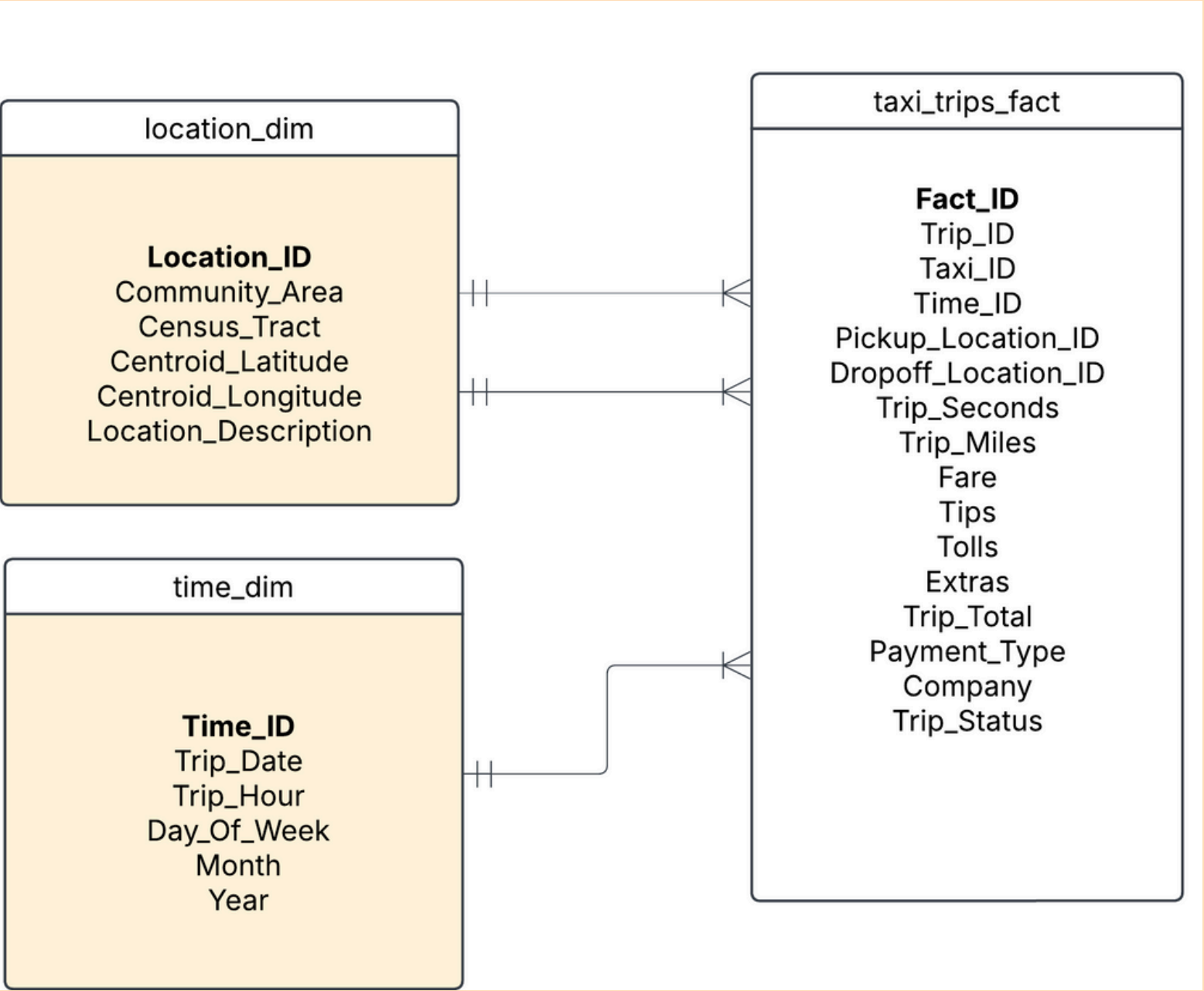
- Insert new locations into location_dim if they don't exist

```
INSERT INTO location_dim (Community_Area, Census_Tract, Centroid_Latitude, Centroid_Longitude)
SELECT DISTINCT Pickup_Community_Area, Pickup_Census_Tract, Pickup_Centroid_Latitude, Pickup_Centroid_Longitude
FROM taxi_trips
WHERE Pickup_Community_Area NOT IN (SELECT Community_Area FROM location_dim);
```

- Insert only new trips into taxi_trips_fact

```
INSERT INTO taxi_trips_fact (
    Trip_ID, Taxi_ID, Time_ID, Pickup_Location_ID, Dropoff_Location_ID,
    Trip_Seconds, Trip_Miles, Fare, Tips, Tolls, Extras, Trip_Total,
    Payment_Type, Company, Trip_Status
)
SELECT
    tt.`Trip ID`,
    tt.`Taxi ID`,
    t.Time_ID,
    pl.Location_ID AS Pickup_Location_ID,
    dl.Location_ID AS Dropoff_Location_ID,
    tt.`Trip Seconds`, tt.`Trip Miles`,
    tt.Fare, tt.Tips, tt.Tolls, tt.Extras, tt.`Trip Total`,
    tt.`Payment Type`, tt.Company, tt.`Trip Status`
FROM taxi_trips tt
JOIN time_dim t
    ON DATE(tt.`Trip Start Timestamp`) = t.Trip_Date
    AND tt.`Trip Start Hour` = t.Trip_Hour
JOIN location_dim pl
    ON tt.`Pickup Community Area` = pl.Community_Area
    AND tt.`Pickup Census Tract` = pl.Census_Tract
JOIN location_dim dl
    ON tt.`Dropoff Community Area` = dl.Community_Area
    AND tt.`Dropoff Census Tract` = dl.Census_Tract
WHERE tt.`Trip Start Timestamp` > (SELECT MAX(t.Trip_Date) FROM taxi_trips_fact f
    JOIN time_dim t ON f.Time_ID = t.Time_ID);
```


STAR SCHEMA



FACT & DIMENSION TABLES

time_dim

Field Name	Type	Description
TIME_ID	INT (PRIMARY KEY)	UNIQUE IDENTIFIER FOR EACH TIME ENTRY (COULD BE YYYYMMDDHH FORMAT)
TRIP_DATE	DATE	DATE OF TRIP
TRIP_HOUR	INTEGER	HOUR OF THE TRIP (0-23)
DAY_OF_WEEK	VARCHAR(10)	NAME OF THE DAY (MONDAY-SUNDAY)
MONTH	INTEGER	MONTH OF THE TRIP (1-12)
YEAR	INTEGER	YEAR OF THE TRIP

location_dim

Field Name	Type	Description
LOCATION_ID	INT (PRIMARY KEY, AUTO-INCREMENT)	PRIMARY KEY (UNIQUE FOR PICKUP & DROPOFF LOCATIONS)
COMMUNITY_AREA	DOUBLE	NUMERIC CODE REPRESENTING THE COMMUNITY AREA IN CHICAGO
CENSUS_TRACT	DOUBLE	CENSUS TRACT ID FOR GEOGRAPHICAL REFERENCE
CENTROID_LATITUDE	DOUBLE	LATITUDE COORDINATE
CENTROID_LONGITUDE	DOUBLE	LONGITUDE COORDINATE
LOCATION_DESCRIPTION	VARCHAR(100)	TEXT DESCRIPTION OF THE LOCATION (E.G., "O'HARE AIRPORT")

taxi_trips_fact

Field Name	Type	Description
FACT_ID	INT (PRIMARY KEY)	UNIQUE IDENTIFIER FOR EACH TRIP RECORD (AUTO-INCREMENTED)
TRIP_ID	VARCHAR(50)	UNIQUE TRIP IDENTIFIER FROM THE TAXI SERVICE
TAXI_ID	VARCHAR(255)	UNIQUE IDENTIFIER FOR THE TAXI VEHICLE
TIME_ID	INT (FOREIGN KEY)	LINKS TO TIME_DIM FOR DATE/TIME-BASED ANALYSIS
PICKUP_LOCATION_ID	INT (FOREIGN KEY)	LINKS TO LOCATION_DIM FOR PICKUP LOCATION
DROPOFF_LOCATION_ID	INT (FOREIGN KEY)	LINKS TO LOCATION_DIM FOR DROPOFF LOCATION
TRIP_SECONDS	DOUBLE	DURATION OF THE TRIP IN SECONDS
TRIP_MILES	DOUBLE	DISTANCE TRAVELED IN MILES
FARE	DOUBLE	BASE FARE CHARGED FOR THE TRIP
TIPS	DOUBLE	TIP AMOUNT PAID BY THE PASSENGER
TOLLS	DOUBLE	TOLL CHARGES FOR THE TRIP
EXTRAS	DOUBLE	ADDITIONAL CHARGES (E.G., AIRPORT FEES)
TRIP_TOTAL	DOUBLE	TOTAL COST OF THE TRIP (SUM OF FARE, EXTRAS, TIPS, AND TOLLS)
PAYMENT_TYPE	VARCHAR(50)	PAYMENT METHOD USED (CASH, CREDIT, MOBILE, ETC.)
COMPANY	VARCHAR(100)	TAXI COMPANY OPERATING THE TRIP
TRIP_STATUS	VARCHAR(50)	INDICATES WHETHER THE TRIP WAS "COMPLETED" OR "CANCELED"

INCREMENTAL UPDATE OF DATA WAREHOUSE

Incremental updates for the Data Warehouse should be scheduled monthly (or daily, depending on the data arrival rate)

Incremental Updates on a Monthly Basis

- Every first day of the month, we run ETL jobs that:
 - **Extract** only new records from taxi_trips
 - **Transform** new timestamps and locations
 - **Load** them into time_dim, location_dim, and taxi_trips_fact
- We use a batch process instead of a full refresh

Scripts/Programs for Automation

- To ensure automated execution, we can use Python and SQLAlchemy to schedule incremental updates and cron jobs

INCREMENTAL UPDATE OF DATA WAREHOUSE

```
import mysql.connector
from datetime import datetime, timedelta

# Database Connection
conn = mysql.connector.connect(
    host="*****",
    user="*****",
    password="*****",
    database="taxi_dw"
)
cursor = conn.cursor()

# Step 1: Get the latest inserted trip date
cursor.execute("""
    SELECT MAX(t.Trip_Date) FROM taxi_trips_fact f
    JOIN time_dim t ON f.Time_ID = t.Time_ID;
""")
last_inserted_date = cursor.fetchone()[0]

# Step 2: Insert new time_dim records
cursor.execute("""
    INSERT INTO time_dim (Trip_Date, Trip_Hour, Day_Of_Week, Month, Year)
    SELECT DISTINCT DATE(Trip_Start_Timestamp), HOUR(Trip_Start_Timestamp),
        DAYNAME(Trip_Start_Timestamp), MONTH(Trip_Start_Timestamp), YEAR(Trip_Start_Timestamp)
    FROM taxi_trips
    WHERE Trip_Start_Timestamp > %s;
""", (last_inserted_date,))
conn.commit()

# Step 3: Insert new location_dim records
cursor.execute("""
    INSERT INTO location_dim (Community_Area, Census_Tract, Centroid_Latitude, Centroid_Longitude)
    SELECT DISTINCT Pickup_Community_Area, Pickup_Census_Tract, Pickup_Centroid_Latitude, Pickup_Centroid_Longitude
    FROM taxi_trips
    WHERE Pickup_Community_Area NOT IN (SELECT Community_Area FROM location_dim);
""")
conn.commit()
```

```
# Step 4: Insert new taxi_trips_fact records
cursor.execute("""
    INSERT INTO taxi_trips_fact (
        Trip_ID, Taxi_ID, Time_ID, Pickup_Location_ID, Dropoff_Location_ID,
        Trip_Seconds, Trip_Miles, Fare, Tips, Tolls, Extras, Trip_Total,
        Payment_Type, Company, Trip_Status
    )
    SELECT
        tt.`Trip ID`,
        tt.`Taxi ID`,
        t.Time_ID,
        pl.Location_ID AS Pickup_Location_ID,
        dl.Location_ID AS Dropoff_Location_ID,
        tt.`Trip Seconds`, tt.`Trip Miles`,
        tt.Fare, tt.Tips, tt.Tolls, tt.Extras, tt.`Trip Total`,
        tt.`Payment Type`, tt.Company, tt.`Trip Status`
    FROM taxi_trips tt
    JOIN time_dim t
        ON DATE(tt.`Trip Start Timestamp`) = t.Trip_Date
        AND tt.`Trip Start Hour` = t.Trip_Hour
    JOIN location_dim pl
        ON tt.`Pickup Community Area` = pl.Community_Area
        AND tt.`Pickup Census Tract` = pl.Census_Tract
    JOIN location_dim dl
        ON tt.`Dropoff Community Area` = dl.Community_Area
        AND tt.`Dropoff Census Tract` = dl.Census_Tract
    WHERE tt.`Trip Start Timestamp` > %s;
""", (last_inserted_date,))
conn.commit()

# Close the connection
cursor.close()
conn.close()
print("Incremental update completed!")
```

Schedule the script to run on the first day of every month at 2 AM:

```
0 2 1 * * /usr/bin/python3 /path/to/taxi_data_update.py
```


OLAP QUERY 1

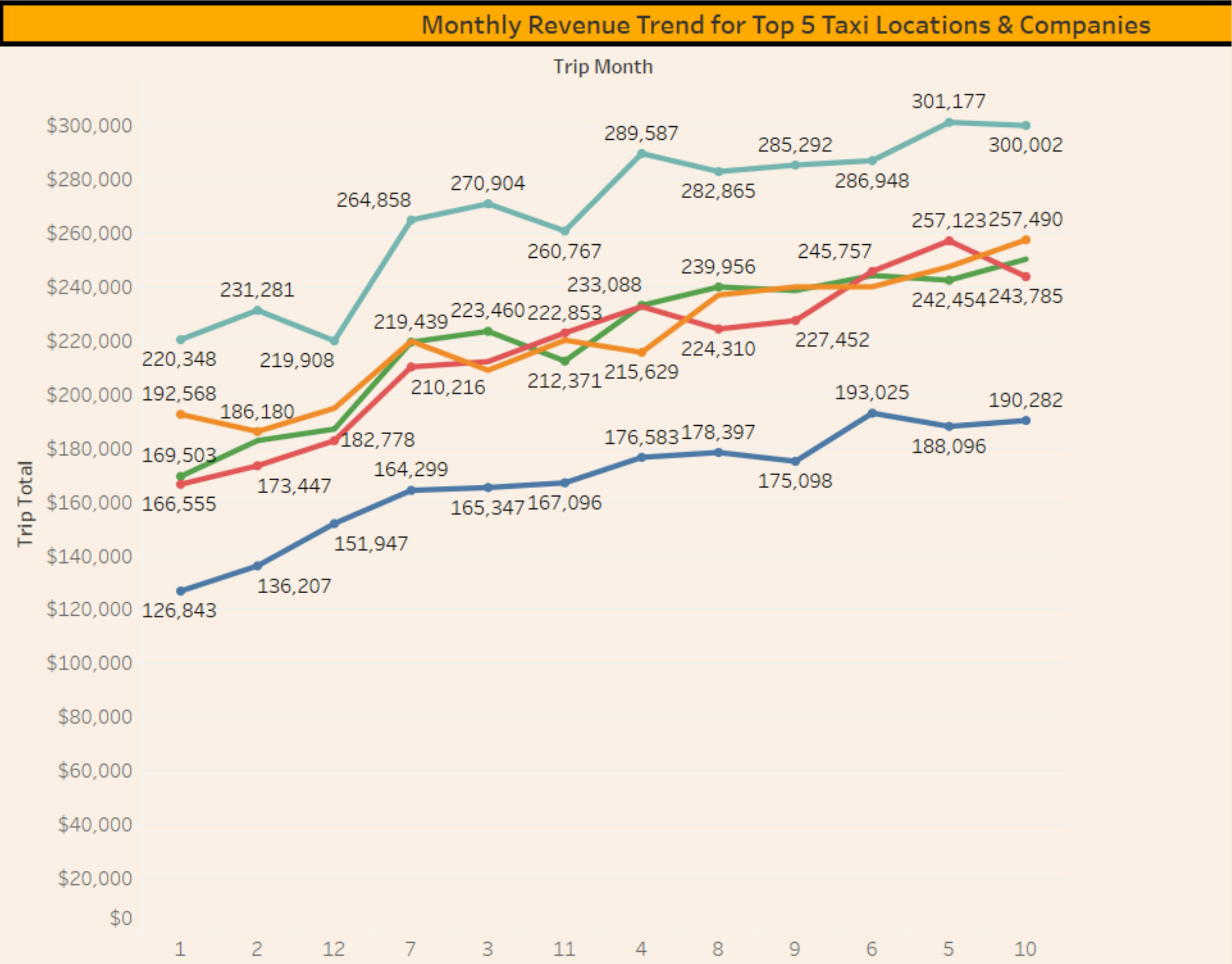
Roll-Up : Monthly Revenue Analysis - For a location (top 5) & Company (Top 5)

```
WITH top_locations AS (  
    SELECT Pickup_Location_ID,  
           SUM(Trip_Total) AS total_revenue  
    FROM taxi_trips_fact  
    GROUP BY Pickup_Location_ID  
    ORDER BY total_revenue DESC  
    LIMIT 5  
,  
top_companies AS (  
    SELECT Company,  
           SUM(Trip_Total) AS total_revenue  
    FROM taxi_trips_fact  
    GROUP BY Company  
    ORDER BY total_revenue DESC  
    LIMIT 5  
)  
SELECT  
    t.Year,  
    t.Month,  
    l.Community_Area AS Location,  
    f.Company,  
    SUM(f.Trip_Total) AS Monthly_Revenue  
FROM taxi_trips_fact f  
JOIN time_dim t ON f.Time_ID = t.Time_ID  
JOIN location_dim l ON f.Pickup_Location_ID = l.Location_ID  
WHERE f.Pickup_Location_ID IN (SELECT Pickup_Location_ID FROM top_locations)  
AND f.Company IN (SELECT Company FROM top_companies)  
GROUP BY t.Year, t.Month, l.Community_Area, f.Company  
ORDER BY t.Year, t.Month, Monthly_Revenue DESC;
```

Result Grid						Filter Rows:	Search	Export:
	Year	Month	Location	Company	Monthly_Revenue			
	2024	1	76	taxi affiliation services	105078.49000000001			
	2024	1	76	sun taxi	94640.95999999989			
	2024	1	76	taxicab insurance agency llc	91526.78000000013			
	2024	1	76	flash cab	70332.38999999993			
	2024	1	76	city service	64070.17			
	2024	1	8	flash cab	39783.71000000003			
	2024	1	8	taxi affiliation services	22065.260000000002			
	2024	1	28	flash cab	20401.040000000015			
	2024	1	32	flash cab	19399.579999999998			
	2024	1	8	sun taxi	15173.169999999987			
	2024	1	8	city service	14764.089999999993			
	2024	1	28	taxi affiliation services	14446.25			
	2024	1	8	taxicab insurance agency llc	12942.979999999996			
	2024	1	32	taxi affiliation services	12918.93			
	2024	1	32	sun taxi	10007.530000000001			
	2024	1	32	taxicab insurance agency llc	9297.26			
	2024	1	32	city service	8311.549999999997			
	2024	1	28	sun taxi	7014.369999999996			

- Aggregates trip revenue at the monthly level, allowing insights into seasonal trends and top-performing areas.
- Uses Common Table Expressions (CTEs) to pre-select the top 5 locations and top 5 companies before aggregation.
- Helps businesses identify high-revenue months and optimize operations in top-performing areas.

VISUAL 1



OLAP QUERY 2

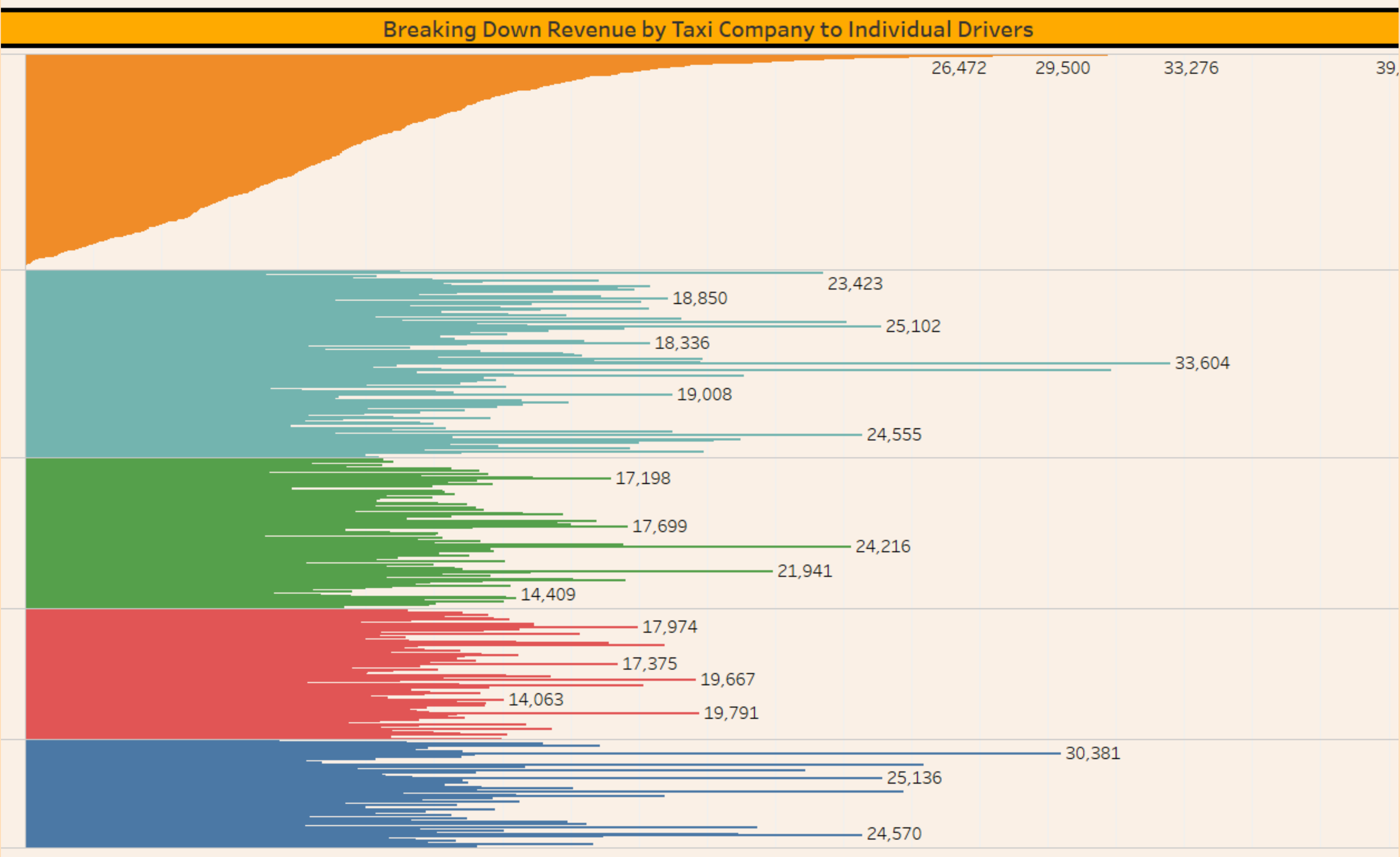
Drill-Down : Breaking Down Revenue by Taxi Company to Individual Drivers

```
SELECT
    f.Company,
    f.Taxi_ID,
    SUM(f.Trip_Total) AS Driver_Revenue
FROM taxi_trips_fact f
GROUP BY f.Company, f.Taxi_ID
ORDER BY f.Company, Driver_Revenue DESC;
```

Result Grid   Filter Rows: Search Export:  Fetch rows: 			
	Company	Taxi_ID	Driver_Revenue
	2733 - 74600 benny jona	5ddeb4816014da6742039f250d1d51b573ba46...	6398.4400000000003
	312 medallion management corp	85369e7993a89ca4df45026bf8e60fbdba04450b...	8132.0800000000001
	312 medallion management corp	95e2243f1be3ddd4b546741f6176fb12f2101fdc3...	7786.8700000000003
	312 medallion management corp	f5d5f5caa931f68cbc6bc434ebe16a90a7fe4a6a...	7107.1499999999997
	312 medallion management corp	63081d550846f1e0c56209bb972eb0fe5b92f5b8...	6387.82
	312 medallion management corp	37da7d7608ce5cab79e71369ed7f4a34f350b4...	6184.4800000000003
	312 medallion management corp	f4ee09a2e3a61fdbfdea39682a992ae964f3387a...	3538.31
	312 medallion management corp	fa265b3601a73684f856bfe63e0b787d9101c3a2...	724.8499999999999
	312 medallion management corp	0090787181662a792d7a7adbc00cf267181d2d5...	174.5
	3556 - 36214 rc andrews cab	de0630d5fd44086c10bf51d621f2151eb38fbc52...	5582.85
	3591 - 63480 chuks cab	38e76b98810925c37414db66db5639e5ddf49c0...	9424.49
	4053 - 40193 adwar h. nikola	275f3f891b8c0ba8f2958873f850d2a03993c99f5...	6469.4200000000003
	5 star taxi	01728d23850d9be83653aa92595ad2a4308e6c...	24783.68999999999...
	5 star taxi	da60c0b2a94e4c516b9c9bdaceeb3f17066f981...	23105.68
	5 star taxi	ee8c4cb9eb10dcb967d64305ca79a9068dbf0b5...	21415.69
	5 star taxi	cb8da8dde1057540d8da485b4510d133316163...	19755.00999999999...
	5 star taxi	f7cc14447361ed368a0904327c2be78af8e9a85...	17680.34
	5 star taxi	c7a8a53874bbcdb11e70a488485e8bdd0bb8cc...	17444.91999999999...

- Starts at the company level and drills down to individual drivers, analyzing their contribution to overall revenue.
- Helps taxi companies identify high-performing drivers and potential training or incentive needs for lower performers.
- Provides insights into fleet utilization and revenue distribution among drivers.

VISUAL 2



OLAP QUERY 3

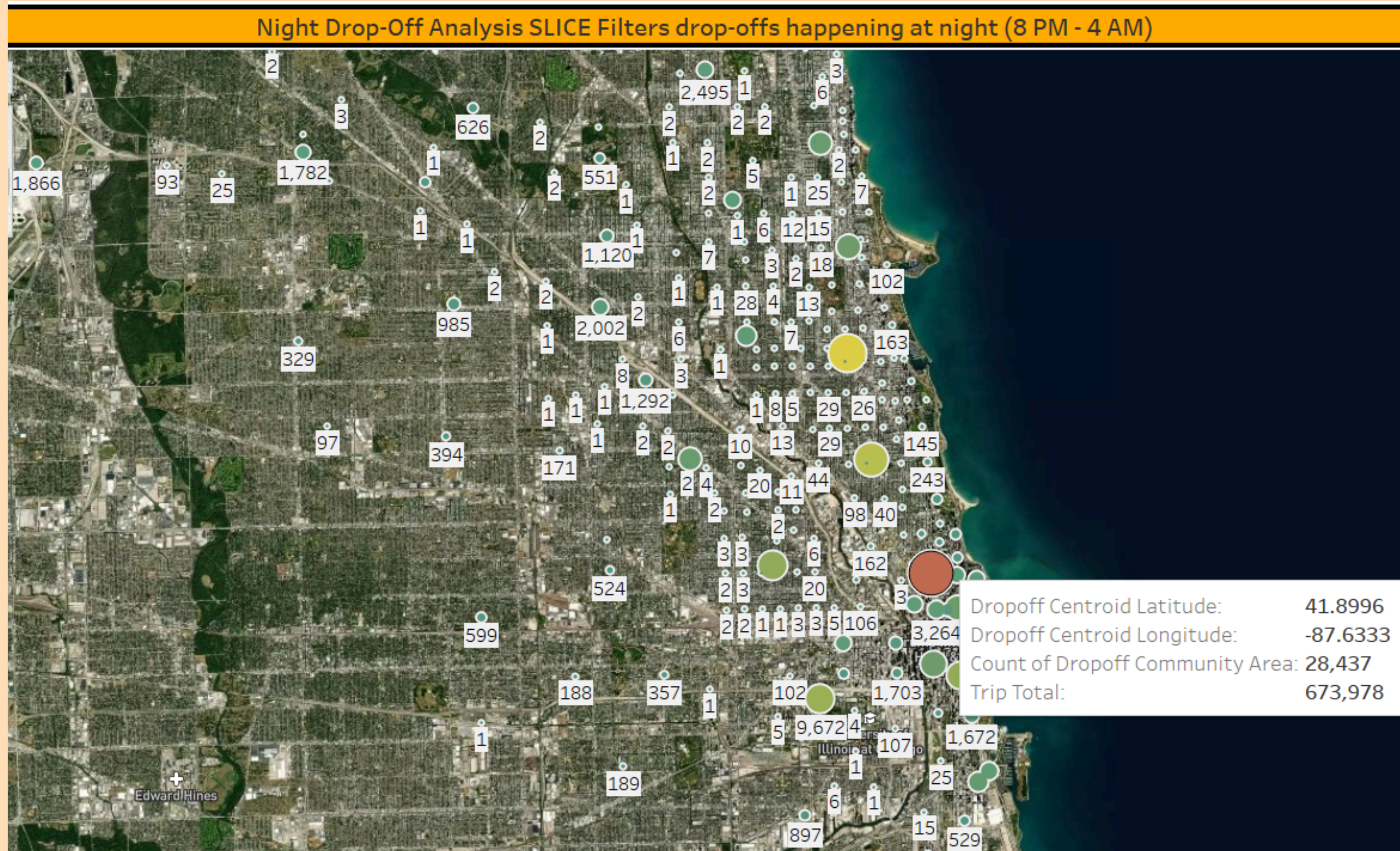
Slice : Night Drop-Off Analysis SLICE Filters
drop-offs happening at night (8 PM - 4 AM)

```
SELECT
  l.Community_Area AS Dropoff_Location,
  COUNT(*) AS Total_Trips,
  SUM(f.Trip_Total) AS Total_Revenue
FROM taxi_trips_fact f
JOIN time_dim t ON f.Time_ID = t.Time_ID
JOIN location_dim l ON f.Dropoff_Location_ID = l.Location_ID
WHERE t.Trip_Hour BETWEEN 20 AND 23 -- 8 PM to 11 PM
      OR t.Trip_Hour BETWEEN 0 AND 4 -- 12 AM to 4 AM
GROUP BY l.Community_Area
ORDER BY Total_Revenue DESC;
```

Result Grid   Filter Rows: Search			
	Dropoff_Locati...	Total_Trips	Total_Revenue
	8	51752	1340443.73000000314
	32	24507	704990.5399999928
	6	14351	424873.8599999974
	28	14644	363522.0499999969
	7	12110	351582.5999999963
	24	8596	262558.0299999994
	76	7641	208680.42000000022
	33	7202	194432.57000000012
	3	5494	150490.99000000002
	22	4227	139363.20000000004
	77	4481	135799.04999999996
	5	2796	100802.68000000007
	1	2248	71420.33000000009
	4	2070	70699.57000000007
	2	2278	64809.81000000003
	16	1879	56861.029999999955
	56	1982	52788.15000000002
	41	1593	50178.89000000002

- Filters trips happening at night, focusing on drop-off locations and total revenue generated.
- Useful for identifying high-demand night-time areas, which can inform safety measures, driver availability, and surge pricing.
- Helps city planners understand late-night travel patterns and improve public transport options.

VISUAL 3



OLAP QUERY 4

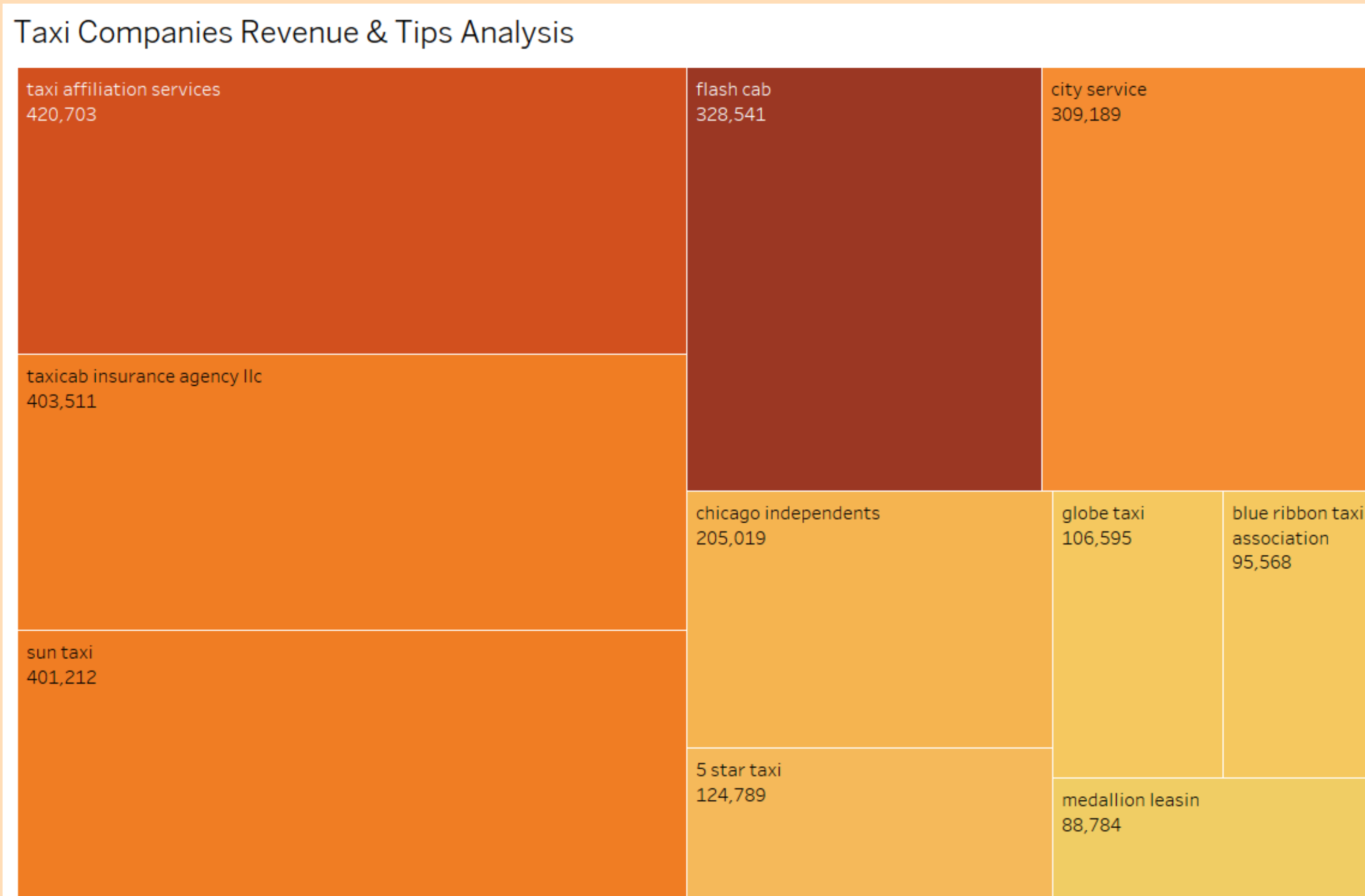
Dice : Average Fare by Hour and Community Areas over a Date Range

```
SELECT
    t.Trip_Hour,
    l.Community_Area AS Pickup_Area,
    AVG(f.Fare) AS Avg_Fare
FROM taxi_trips_fact f
JOIN time_dim t ON f.Time_ID = t.Time_ID
JOIN location_dim l ON f.Pickup_Location_ID = l.Location_ID
WHERE t.Trip_Date BETWEEN '2024-04-01' AND '2024-04-30' -- Date Range
AND l.Community_Area IN (8, 28, 32) -- Specific Community Areas
GROUP BY t.Trip_Hour, l.Community_Area
ORDER BY t.Trip_Hour, l.Community_Area;
```

Result Grid   Filter Rows: Search			
	Trip_Hour	Pickup_Area	Avg_Fare
	0	8	13.613999999999999
	0	28	17.13595041322314
	0	32	17.62232558139535
	1	8	13.455624999999996
	1	28	15.716582278481015
	1	32	16.343333333333333
	2	8	13.229103448275865
	2	28	16.111176470588237
	2	32	15.539523809523809
	3	8	15.882608695652172
	3	28	23.19615384615384
	3	32	21.93518518518518
	4	8	22.430384615384614
	4	28	27.63826086956522
	4	32	30.75547169811321
	5	8	29.045555555555556
	5	28	15.442
	5	32	31.984248366013073

- Filters by multiple dimensions (time range and specific community areas) and analyzes fare trends by the hour.
- Helps determine peak fare hours, allowing dynamic pricing adjustments to maximize revenue.
- Useful for demand forecasting, helping taxi companies allocate cars efficiently across time and locations.

VISUAL 4



OLAP QUERY 5

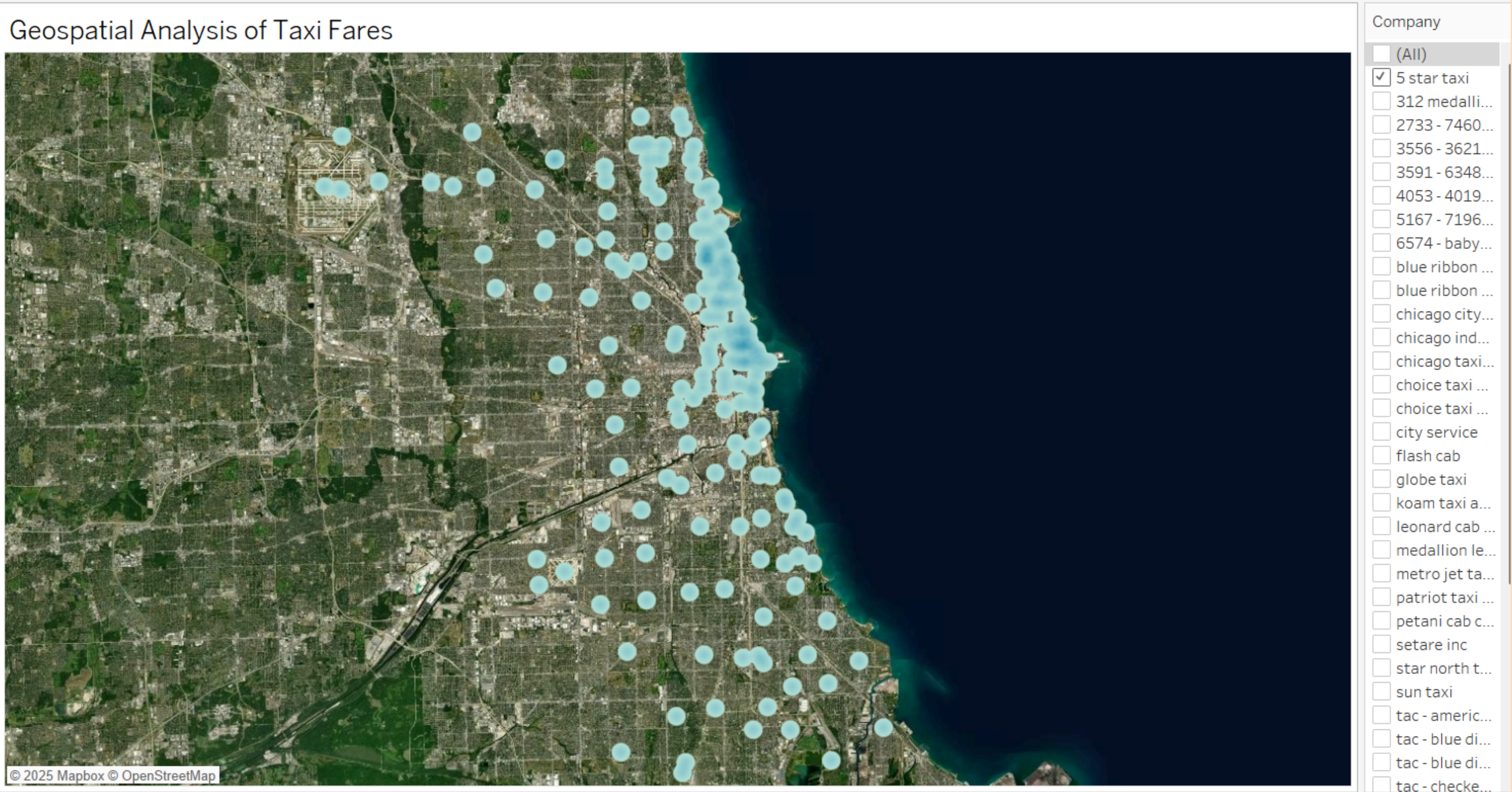
Pivot : Tips vs. Fare Amount by Passenger Count

```
SELECT
  t.Trip_Hour,
  SUM(f.Tips) AS Total_Tips,
  SUM(f.Fare) AS Total_Fare,
  ROUND((SUM(f.Tips) / SUM(f.Fare)) * 100, 2) AS Tip_Percentage
FROM taxi_trips_fact f
JOIN time_dim t ON f.Time_ID = t.Time_ID
GROUP BY t.Trip_Hour
ORDER BY t.Trip_Hour;
```

Result Grid   Filter Rows: Search Export: 				
	Trip_Hour	Total_Tips	Total_Fare	Tip_Percentage
	0	55980.25000000046	420839.14999999994	13.3
	1	29435.54999999993	230711.53000000001	12.76
	2	13467.819999999989	119946.72999999997	11.23
	3	9793.080000000016	103549.06000000001	9.46
	4	15006.799999999972	162044.66000000003	9.26
	5	34505.789999999855	310759.48999999999	11.1
	6	56303.830000000366	510289.35999999991	11.03
	7	86369.17000000003	722880.34999999986	11.95
	8	118803.10000000126	947500.1800000014	12.54
	9	138710.73000000158	1135330.2300000023	12.22
	10	137219.20000000278	1218549.3300000017	11.26
	11	150816.36000000234	1338733.3400000054	11.27
	12	162553.9800000001	1403421.6200000022	11.58
	13	173470.60000000193	1440309.0899999985	12.04
	14	171308.26000000042	1432233.4000000004	11.96
	15	164433.33000000136	1373509.2300000044	11.97
	16	165515.2300000015	1312616.2000000016	12.61
	17	178541.16999999937	1311611.8500000027	13.61

- Analyzes tipping trends across different hours of the day, helping understand customer behavior.
- Provides insights into which time periods generate higher tips relative to fare amount, potentially influencing driver shift planning.
- Useful for driver training programs, as they can learn when customers are more likely to tip.

VISUAL 5

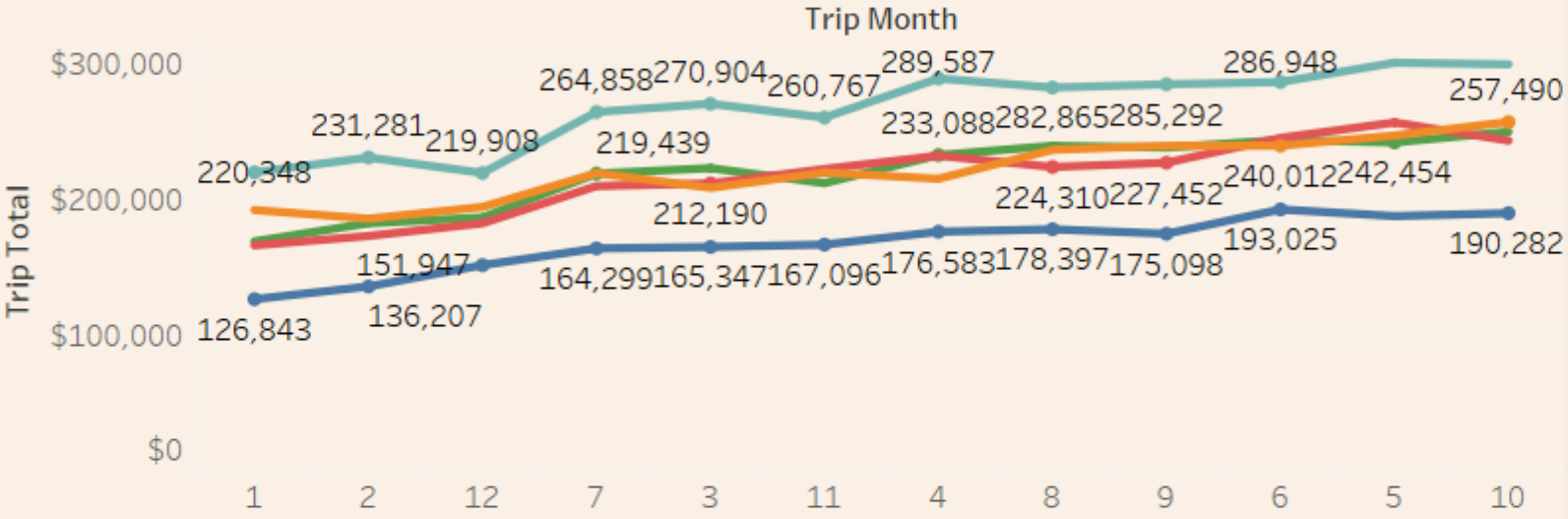


DASHBOARD

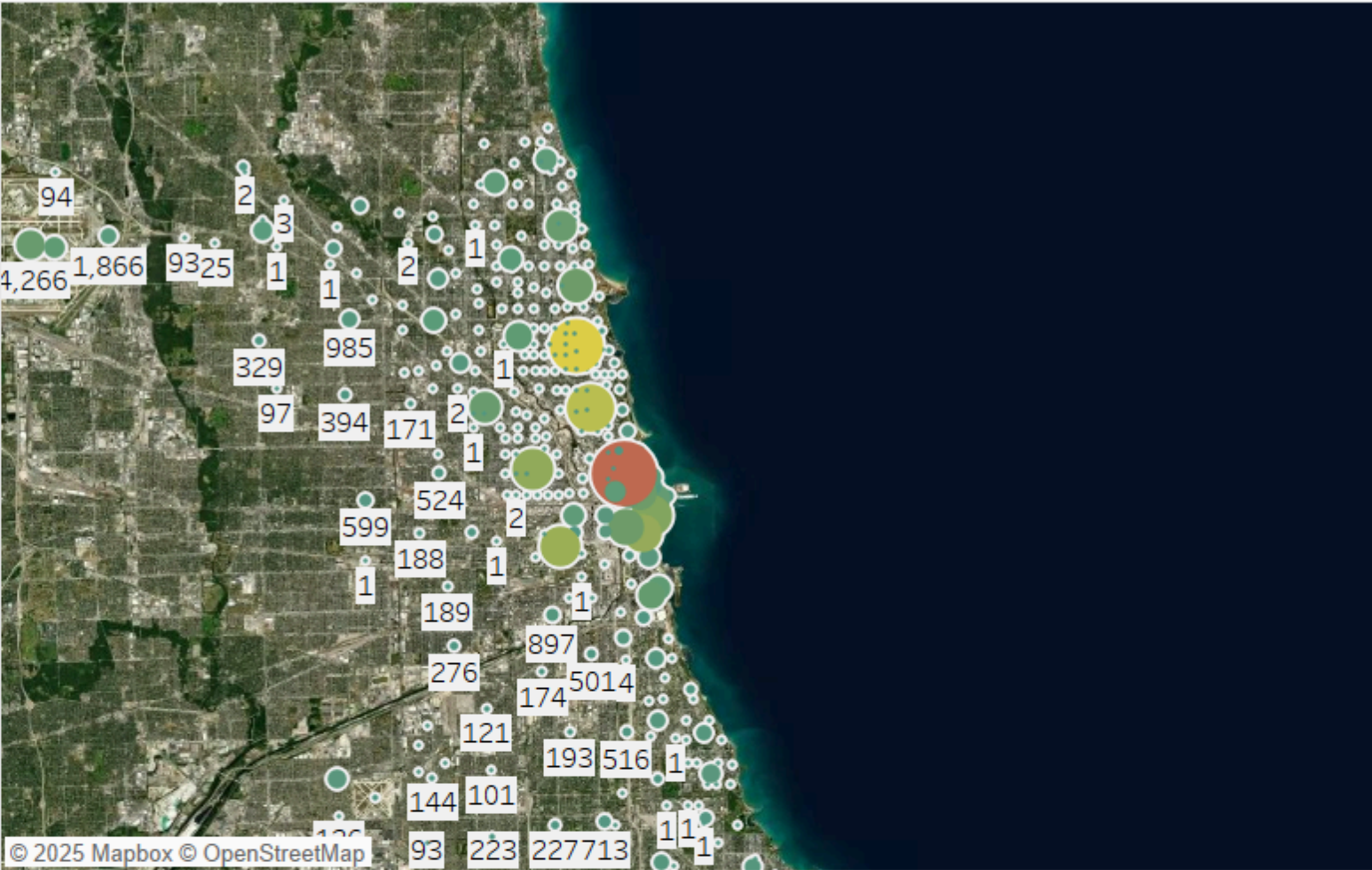
Total Taxi Trips in Chicago

11,05,495

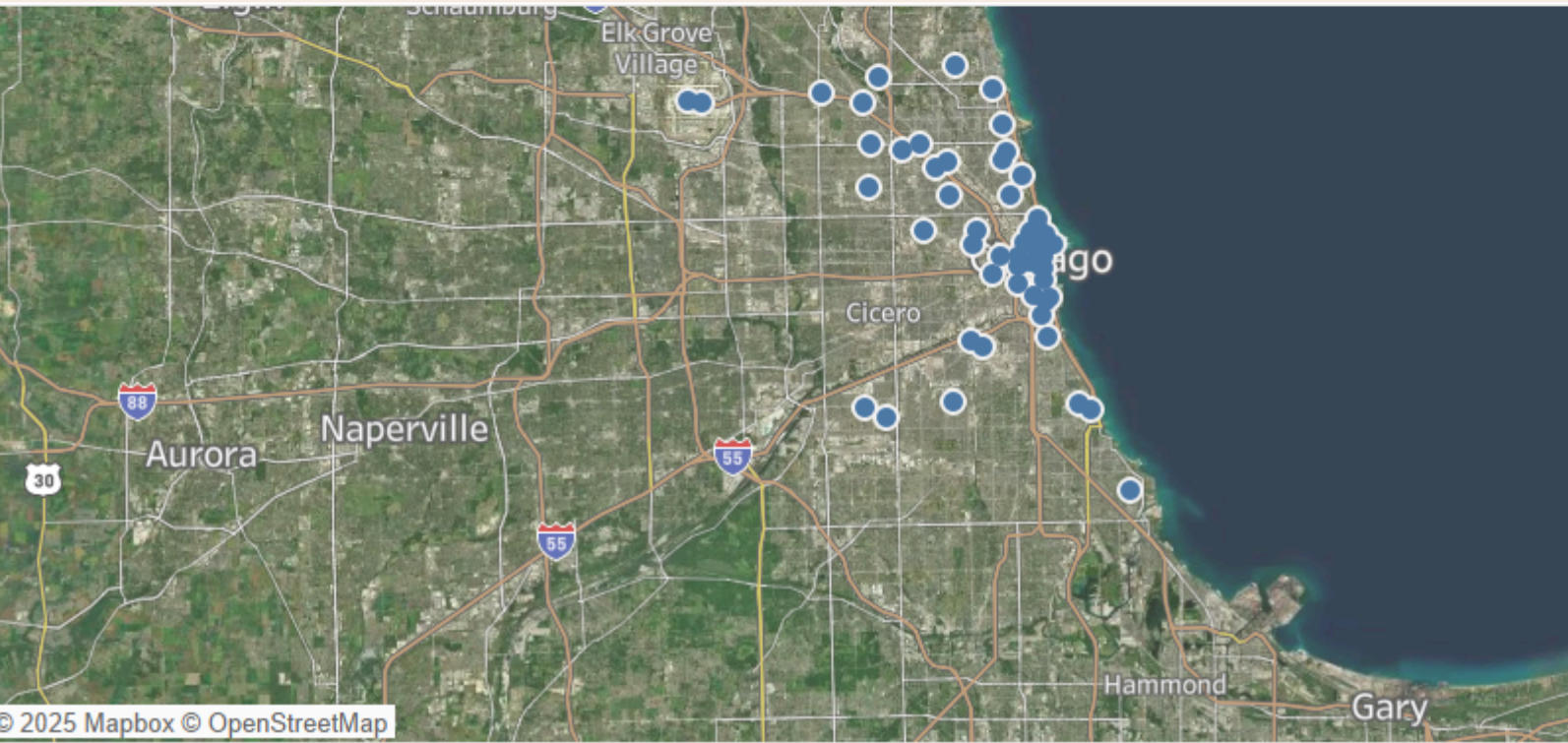
Monthly Revenue Trend for Top 5 Taxi Locations & Companies



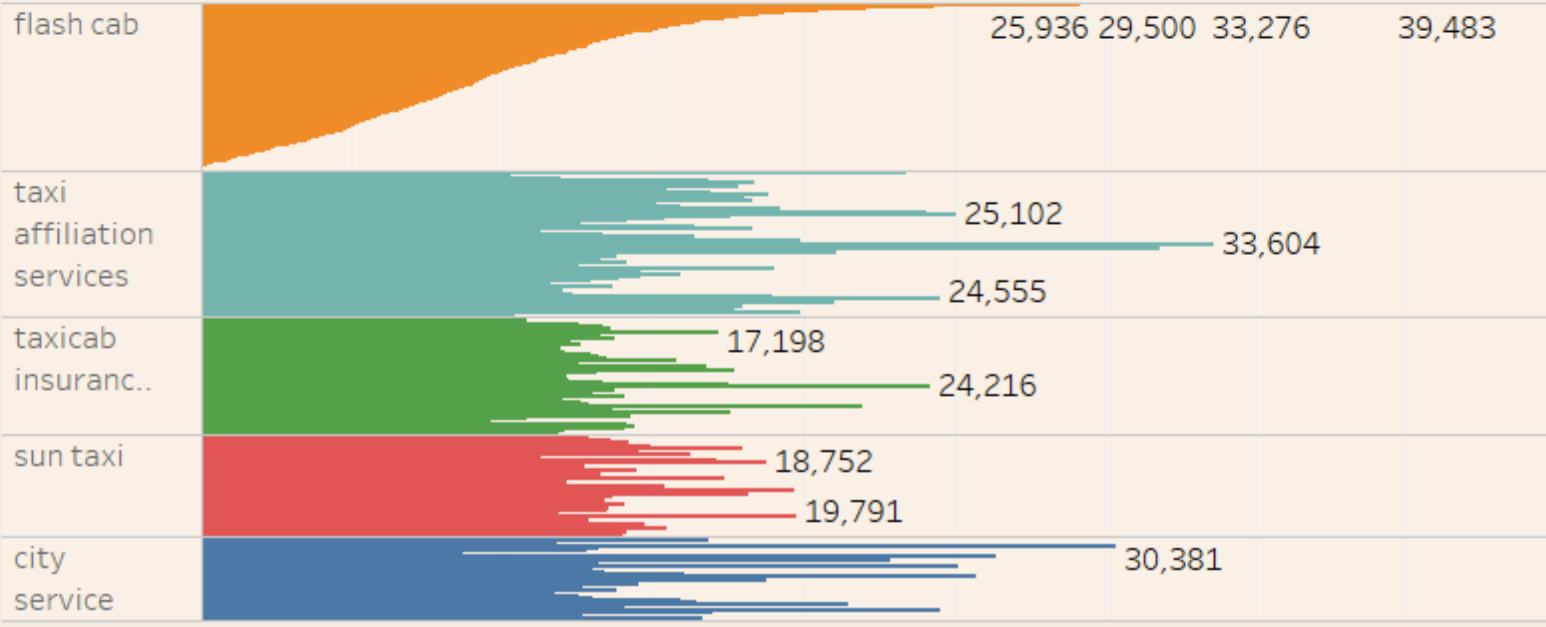
Night Drop-Off Analysis SLICE Filters drop-offs happening at night (8 PM - 4 AM)



Geospatial Analysis of Taxi Fares



Breaking Down Revenue by Taxi Company to Individual Drivers



CONCLUSION

Challenges

- Ensuring data accuracy across sources.
- Optimizing query performance for large datasets.
- Handling data integrity and cleaning issues.

Learnings

- Importance of data validation and cleaning.
- Recognizing trip patterns and peak demand.
- Effective geospatial analysis for insights.

Improvements

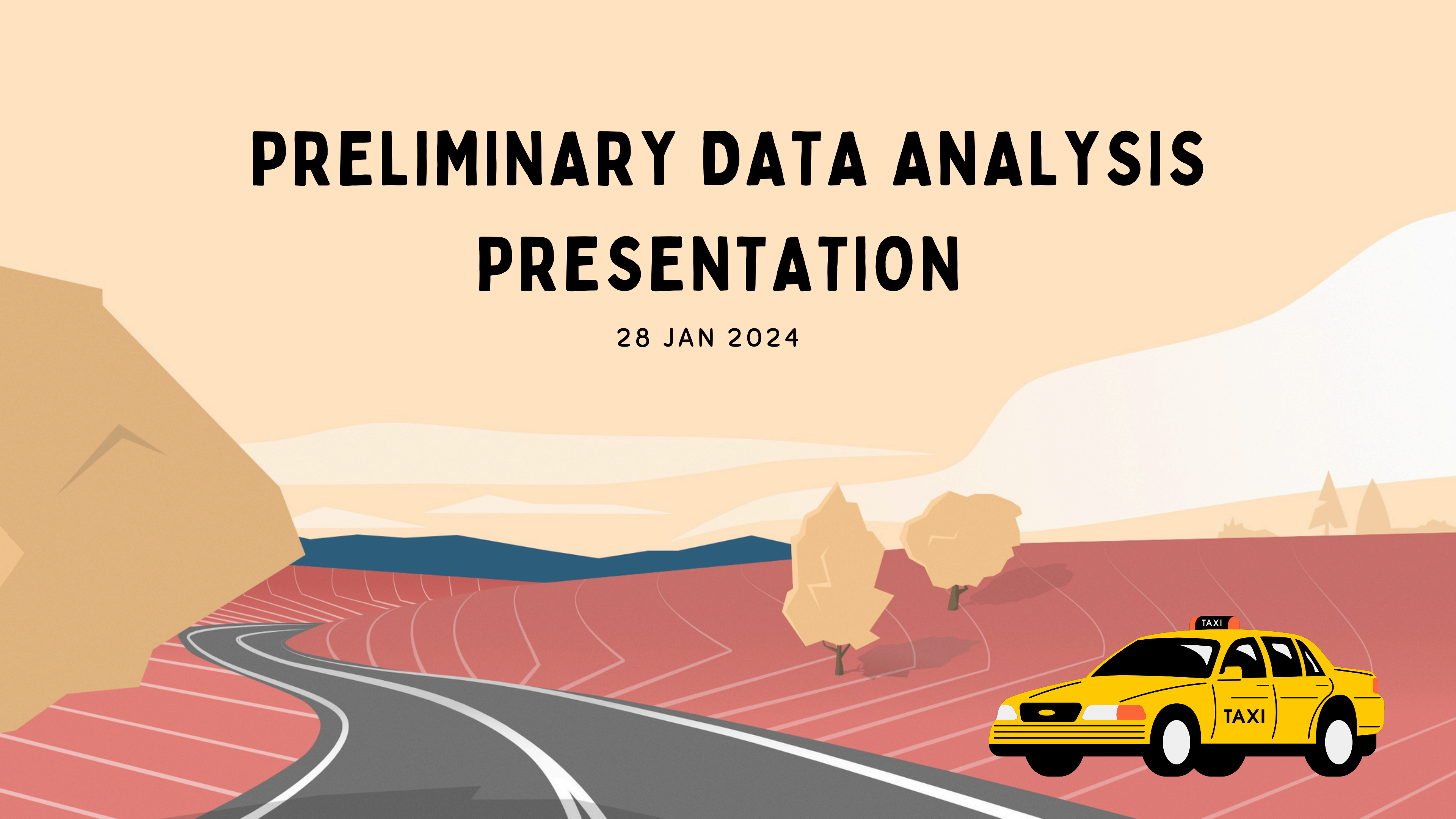
- Integrate real-time data for better accuracy.
- Enhance geospatial visualizations for clarity.
- Optimize queries for improved performance.

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THANK YOU
ANY QUESTIONS?

PRELIMINARY DATA ANALYSIS PRESENTATION

28 JAN 2024



PURPOSE OF DATA COLLECTION

Key Insights:

- **Operational Efficiency:** Trip durations, distances, routes, and peak demand.
- **Customer Behavior:** Payment preferences and tipping patterns.
- **Geographic Insights:** Popular pickup/drop-off locations and community-level trip analysis.
- **Revenue Optimization:** Anomalies in fare structures and additional charges.
- **Data Quality:** Addressing missing or anomalous data.

Objective:

1. Analyze and optimize taxi services in Chicago.
2. Understand transportation patterns, operational efficiencies, and passenger behaviors.

DATA COLLECTION

01

Source: Taxi meters integrated with GPS and point-of-sale systems.

02

Sampling: Census-style collection of all taxi trips during 2024.

03

Timeframe: Includes both AM and PM trips for the year 2024.

04

Demographics:

- Focus on Chicago (community areas, census tracts, geospatial data).
- Indirect passenger insights through payment modes and trip locations.



DATA CLEANING

01

Fill missing numeric values with the median

'Trip Seconds', 'Trip Miles', 'Trip Total', 'Extras', 'Tips', 'Tolls', 'Fare'

02

Fill missing categorical values with 'Unknown'

'Payment Type', 'Company'

03

**Drop rows with missing critical location values
(latitudes/longitudes)**

'Pickup Centroid Latitude', 'Pickup Centroid Longitude', 'Dropoff Centroid Latitude', 'Dropoff Centroid Longitude'



OUTLIER REMOVAL USING IQR METHOD

01

Outlier Detection: Identified outliers in key numeric columns ('Trip Miles', 'Trip Seconds', 'Trip Total') using the Interquartile Range (IQR) method.

02

Methodology:

- Calculated the first (Q1) and third (Q3) quartiles.
- Defined outliers as values outside the range:
 - Lower Bound = $Q1 - 1.5 \times IQR$
 - Upper Bound = $Q3 + 1.5 \times IQR$
- Filtered out rows where values were below the lower bound or above the upper bound.

03

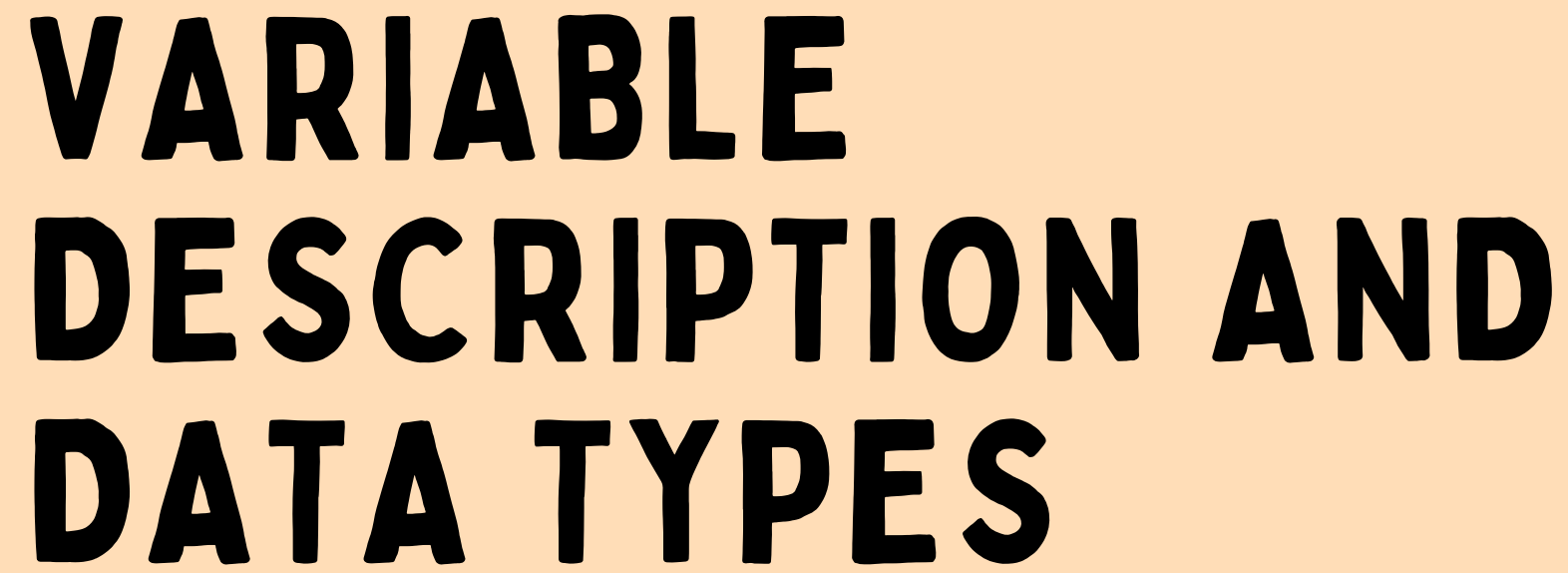
Result: Cleaned Dataset: Outliers were successfully removed, resulting in a more reliable dataset for analysis.



DESCRIPTIVE STATISTICS

	Trip Seconds	Trip Miles	Fare	Tips	Trip Total
count	647934.000000	648028.000000	646274.000000	646274.000000	646274.000000
mean	1272.107150	6.810736	23.037352	2.964785	28.440445
std	1618.118758	7.983354	30.334046	4.321618	36.354903
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	480.000000	1.000000	8.500000	0.000000	10.500000
50%	959.000000	3.300000	16.250000	1.000000	19.500000
75%	1743.000000	12.050000	35.000000	4.210000	44.100000
max	85829.000000	2166.390000	8050.000000	297.000000	8893.380000

- 1. **Trip Duration (Seconds):** The average trip lasts about 21 minutes (1272 seconds), with a wide range, from 0 to over 23 hours.
- 2. **Trip Distance (Miles):** The average trip covers 6.81 miles, with a range from 0 to over 2166 miles (likely due to data issues or outliers).
- 3. **Fare:** The average fare is \$23.04, but there's significant variation (std = \$30.33), with fares ranging from \$0 to \$8050.
- 4. **Tips:** The average tip is \$2.96, with some trips receiving high tips up to \$297, indicating potential outliers or special cases.
- 5. **Total Trip Cost:** The total cost, combining fare and tips, averages \$28.44, with a broad range from \$0 to \$8893.38.
- 6. **Outliers:** There are significant outliers, especially in trip distance, fare, and total cost, suggesting data cleaning might be needed.



```
[5 rows x 23 columns]
```




VARIABLE DESCRIPTION AND DATA TYPES

Variable	Data Type	Type	Description
Trip ID	string	Categorical	Unique identifier for each trip
Taxi ID	string	Categorical	Unique identifier for each taxi
Trip Start Timestamp	datetime	Categorical	The start time of the trip
Trip End Timestamp	datetime	Categorical	The end time of the trip
Trip Seconds	float	Numerical	Duration of the trip in seconds
Trip Miles	float	Numerical	Distance covered in miles



VARIABLE DESCRIPTION AND DATA TYPES

Variable	Data Type	Type	Description
Pickup Census Tract	float	Numerical	Census identifier for the pickup location
Dropoff Census Tract	float	Numerical	Census identifier for the dropoff location
Pickup Community Area	float	Numerical	Community ID for the pickup area
Dropoff Community Area	float	Numerical	Community ID for the dropoff area
Fare	float	Numerical	Fare charged for the trip
Tips	float	Numerical	Tips given for the trip



VARIABLE DESCRIPTION AND DATA TYPES

Variable	Data Type	Type	Description
Tolls	float	Numerical	Tolls incurred during the trip
Extras	float	Numerical	Additional charges (e.g., waiting time)
Trip Total	float	Numerical	Total fare, including tips, tolls, and extras
Payment Type	string	Categorical	The mode of payment (e.g., cash, credit card)
Company	string	Categorical	The taxi company managing the trip
Pickup Centroid Latitude	float	Numerical	Latitude of the pickup location



VARIABLE DESCRIPTION AND DATA TYPES

Variable	Data Type	Type	Description
Pickup Centroid Longitude	float	Numerical	Longitude of the pickup location
Pickup Centroid Location	string	Categorical	Geospatial representation of the pickup location (e.g., "POINT (x y)")
Dropoff Centroid Latitude	float	Numerical	Latitude of the dropoff location
Dropoff Centroid Longitude	float	Numerical	Longitude of the dropoff location
Dropoff Centroid Location	string	Categorical	Geospatial representation of the dropoff location (e.g., "POINT (x y)")



EXPLANATION OF DATA TYPES

Categorical Variables:

- **Definition:** These variables describe categories or groups, often represented as strings or IDs
- **Examples:** Trip ID, Taxi ID, Payment Type, Company, Pickup Centroid Location, Dropoff Centroid Location

Numerical Variables:

- **Definition:** These variables contain measurable values, either continuous (e.g., Trip Miles, Fare) or discrete (e.g., Pickup Community Area if converted to integers)
- **Examples:** Trip Seconds, Trip Miles, Fare, Tips, Tolls, Extras, Trip Total, Pickup Centroid Latitude, Pickup Centroid Longitude



EXPLANATION OF DATA TYPES

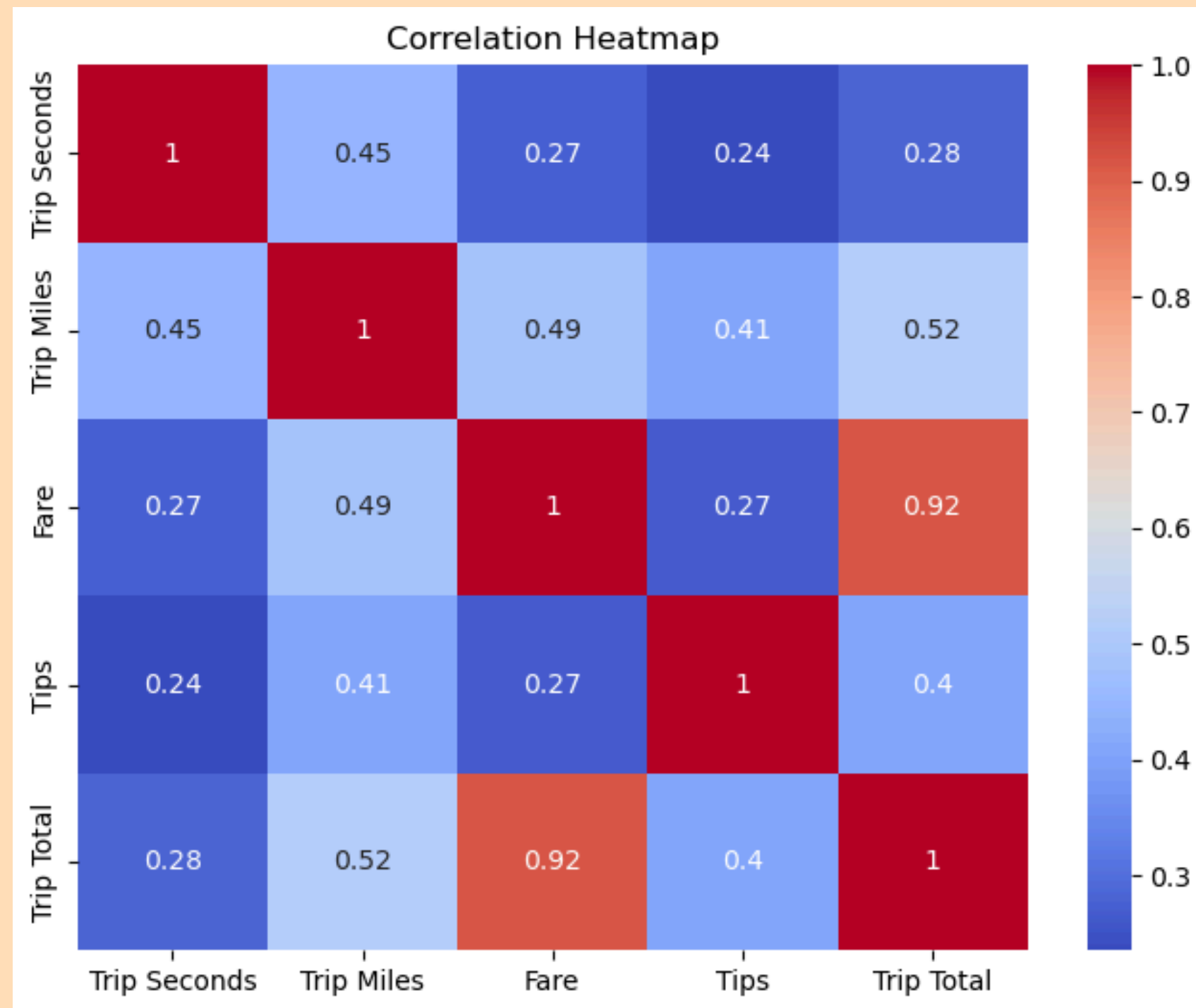
Datetime Variables:

- **Definition:** These variables represent date and time, useful for time-based operations
- **Examples:** Trip Start Timestamp, Trip End Timestamp

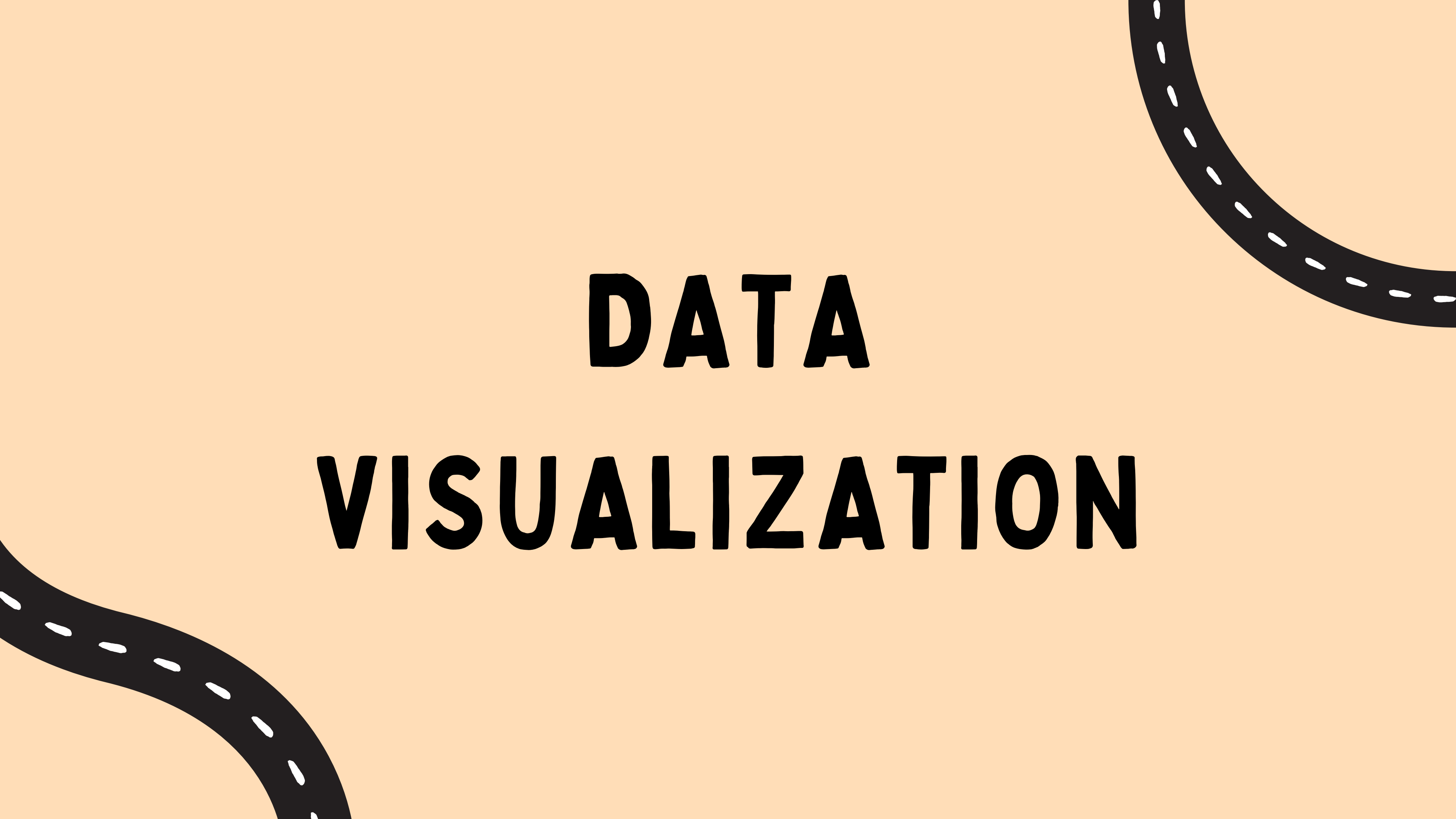
Special Cases:

- **Community Area IDs:** These can be stored as integers (int) if there are no missing values, but remain float to account for potential gaps in the dataset.
- **Geospatial Locations:** While latitude and longitude are float, the geospatial points (Pickup Centroid Location, Dropoff Centroid Location) are stored as strings

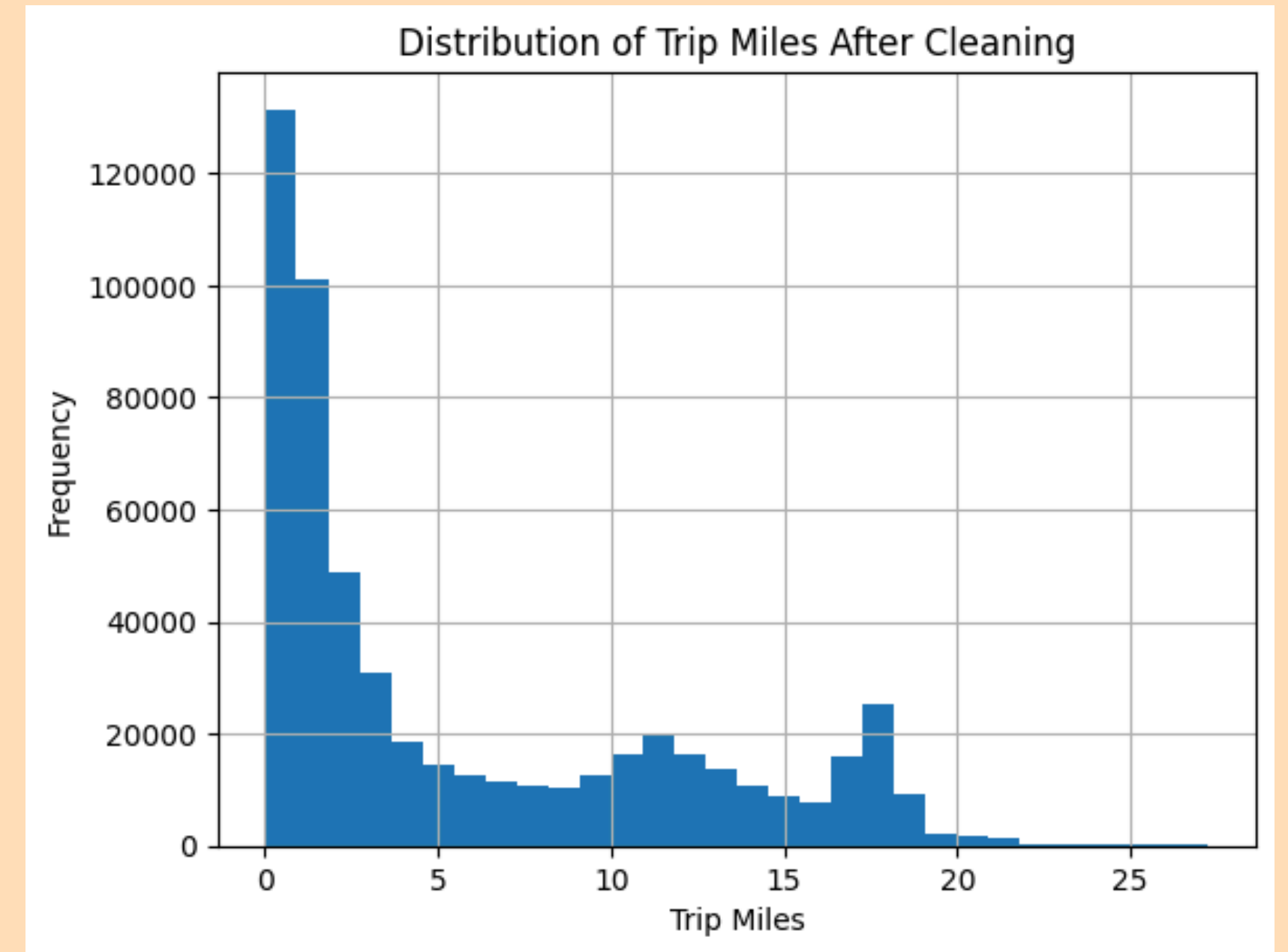
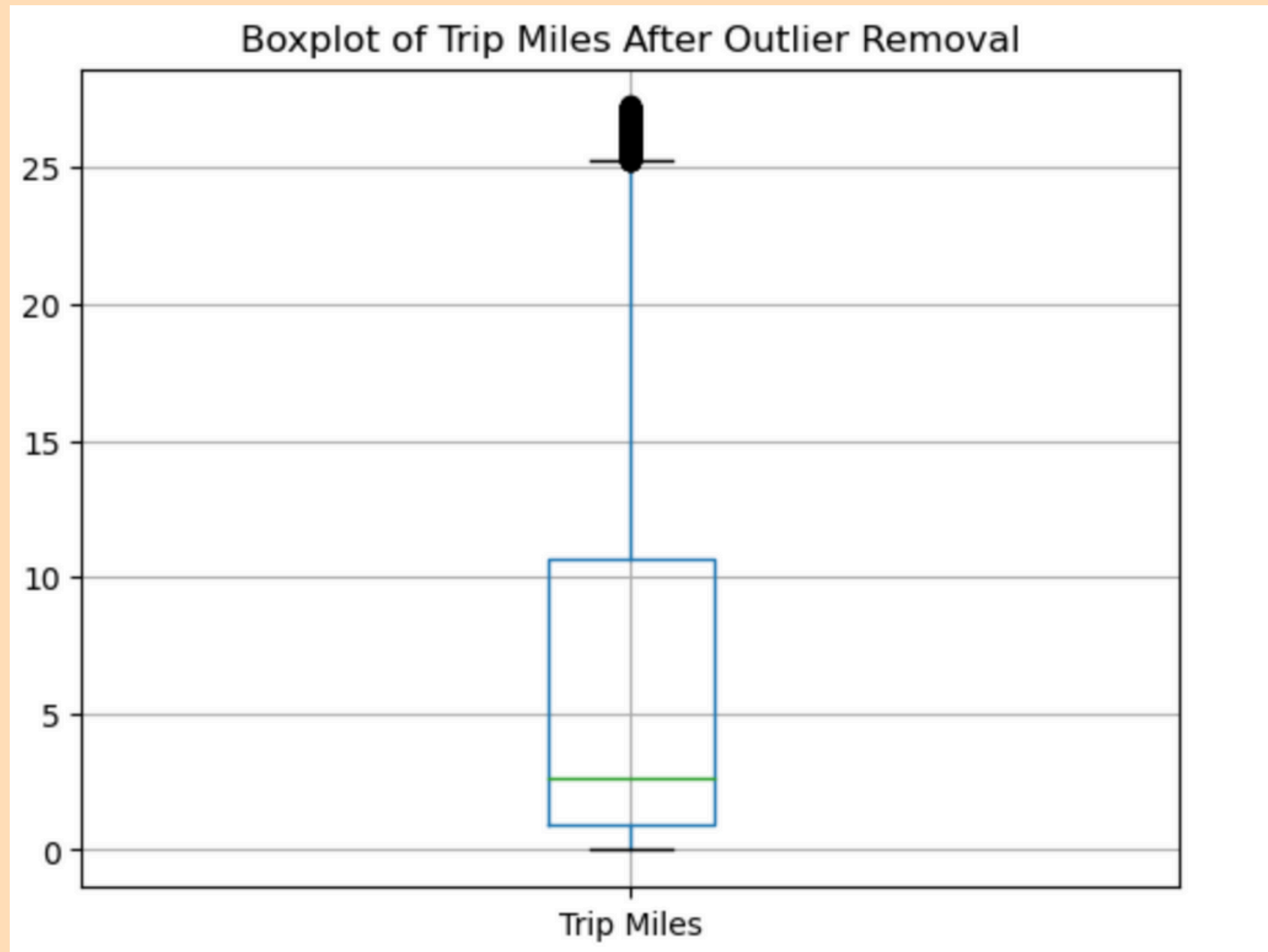
Correlation Analysis



- **Fare and Trip Total (0.92):** Strong correlation, as trip total is mainly based on fare, plus tips and tolls.
- **Trip Miles and Fare (0.49):** Moderate correlation, indicating longer trips usually cost more, but other factors like base fare affect the price.
- **Trip Miles and Trip Total (0.52):** Moderate correlation, with distance impacting total cost, though other factors (e.g., tolls) also play a role.
- **Trip Seconds and Trip Miles (0.45):** Moderate correlation, suggesting longer trips usually take more time, but traffic can vary the relationship.
- **Tips and Other Variables (~0.4):** Weak correlation, indicating tips are influenced more by service or personal factors than by fare or trip length.

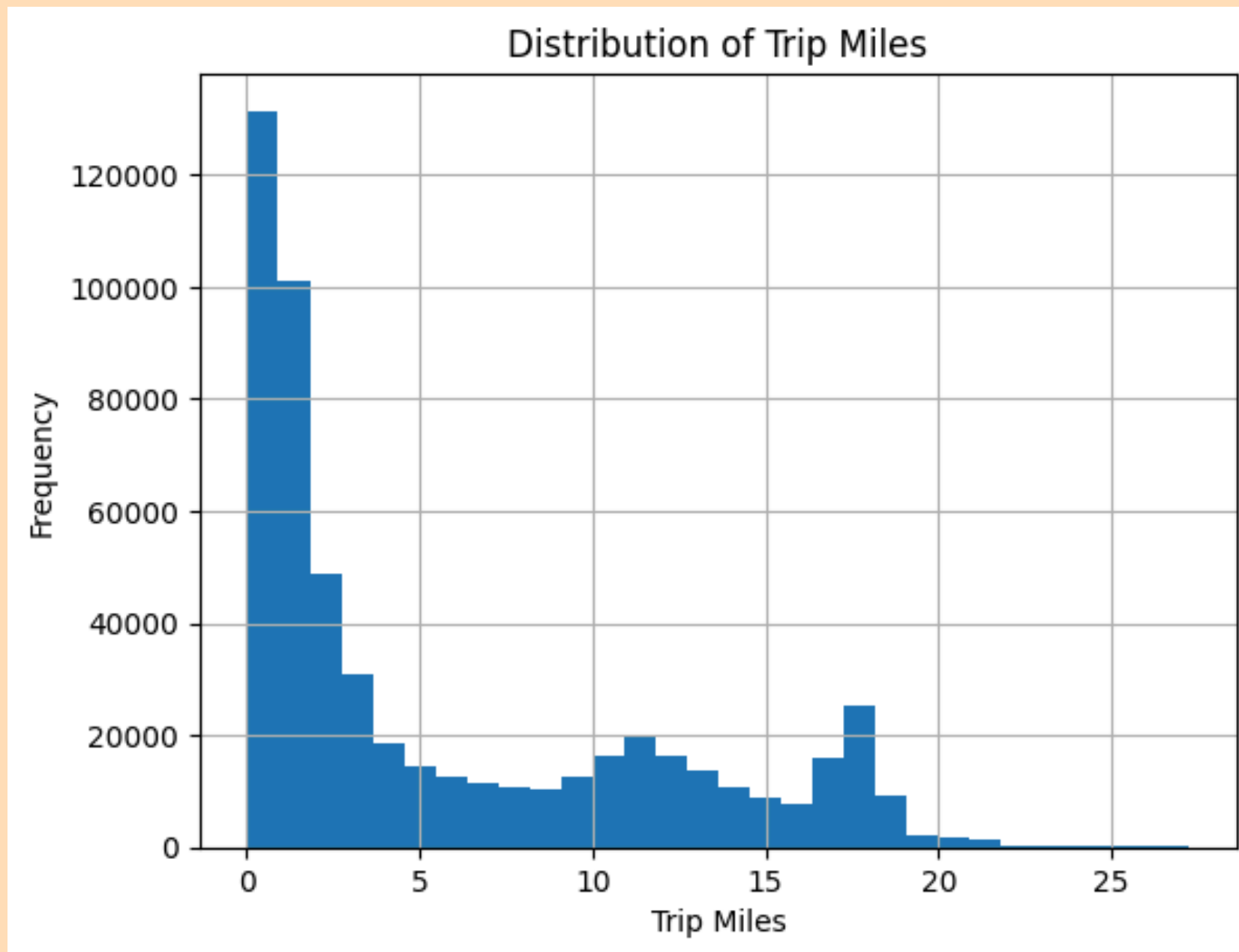
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DATA VISUALIZATION

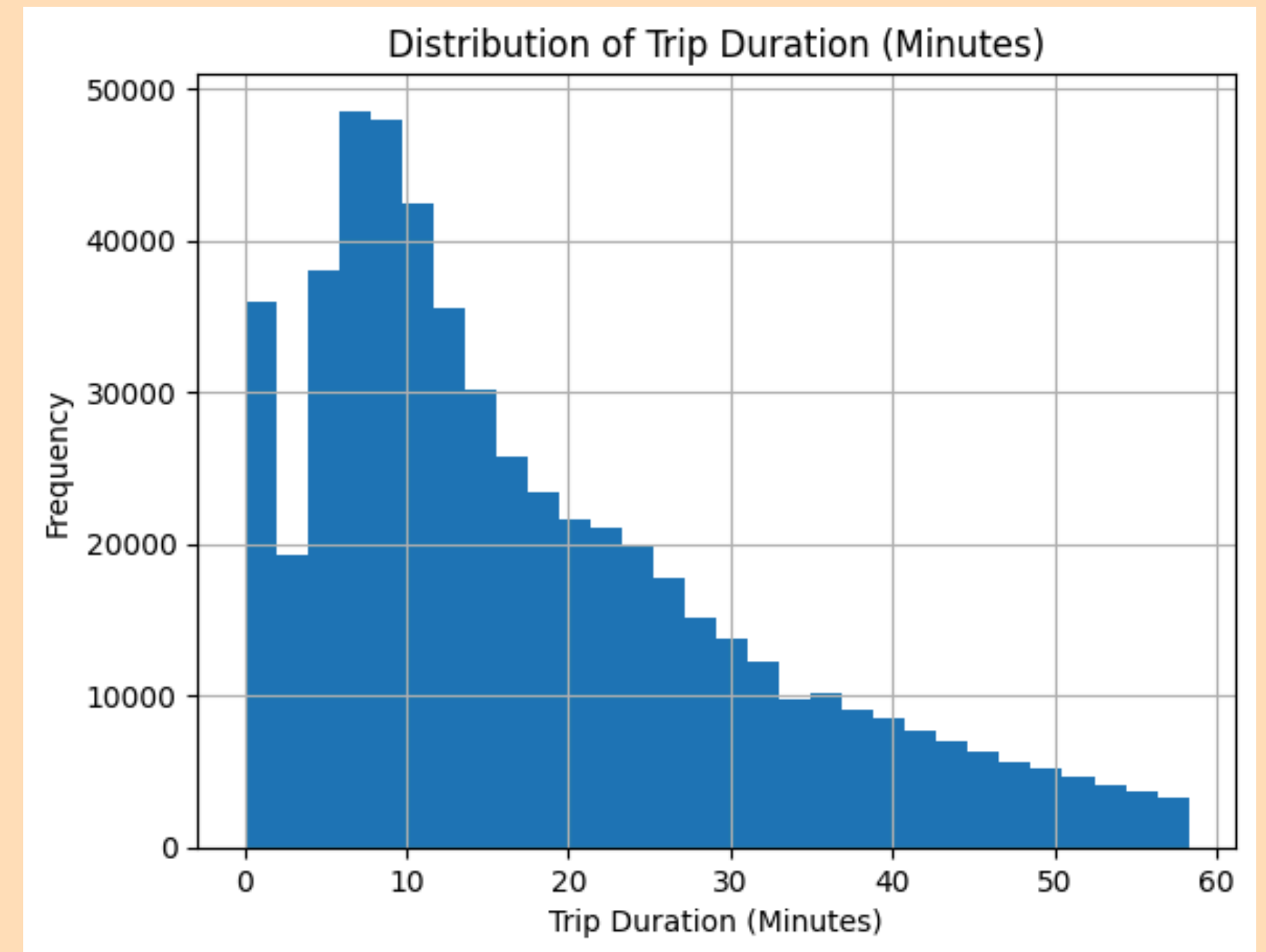


- Boxplot shows "Trip Miles" distribution after outlier removal, with median and IQR.
- No extreme values, indicating a cleaner dataset.

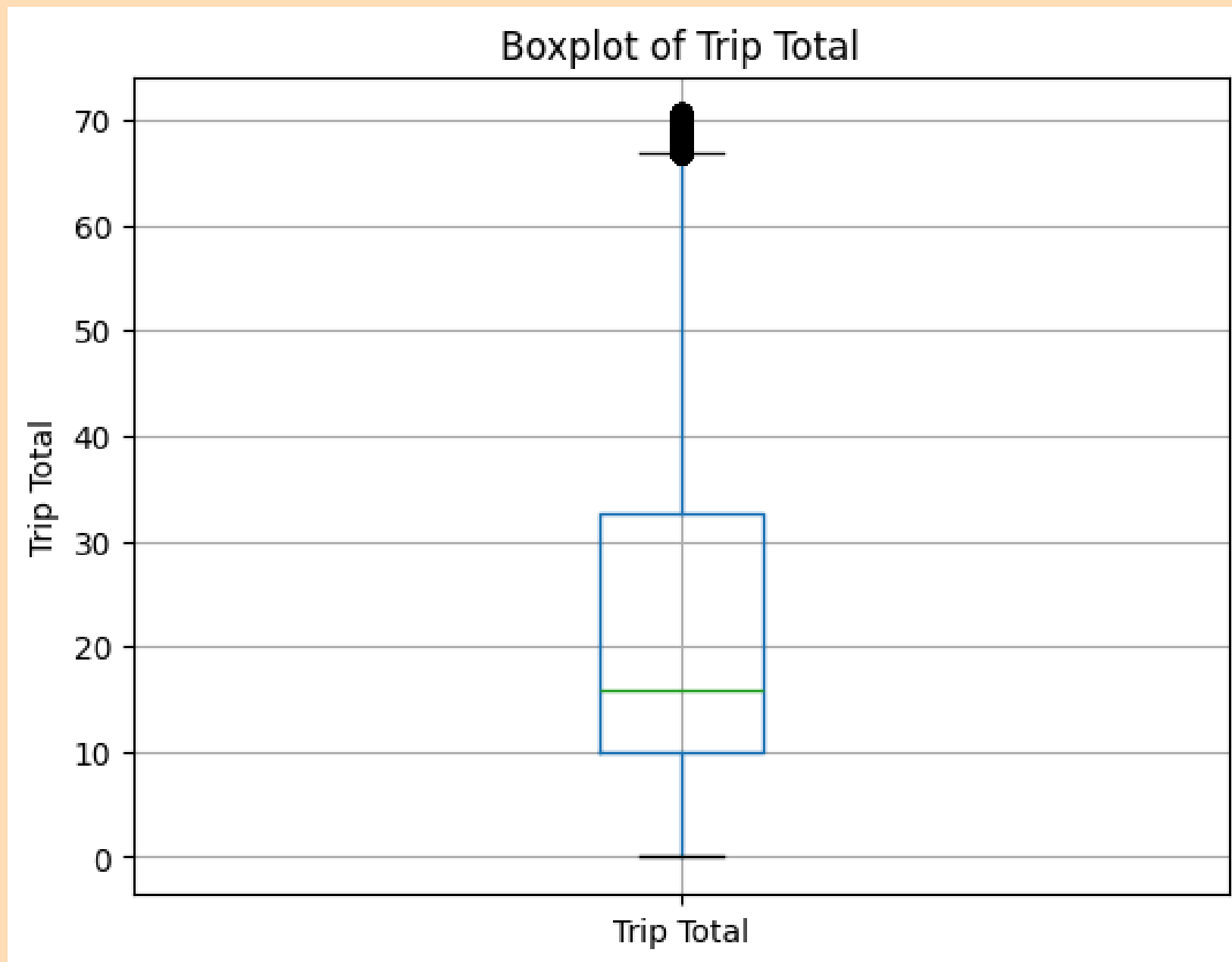
- Most trips fall between 0 and 5 miles, with fewer longer trips, reflecting typical city taxi patterns.
- Outliers (e.g., extremely long trips) were removed, retaining valid data up to 27.25 miles.



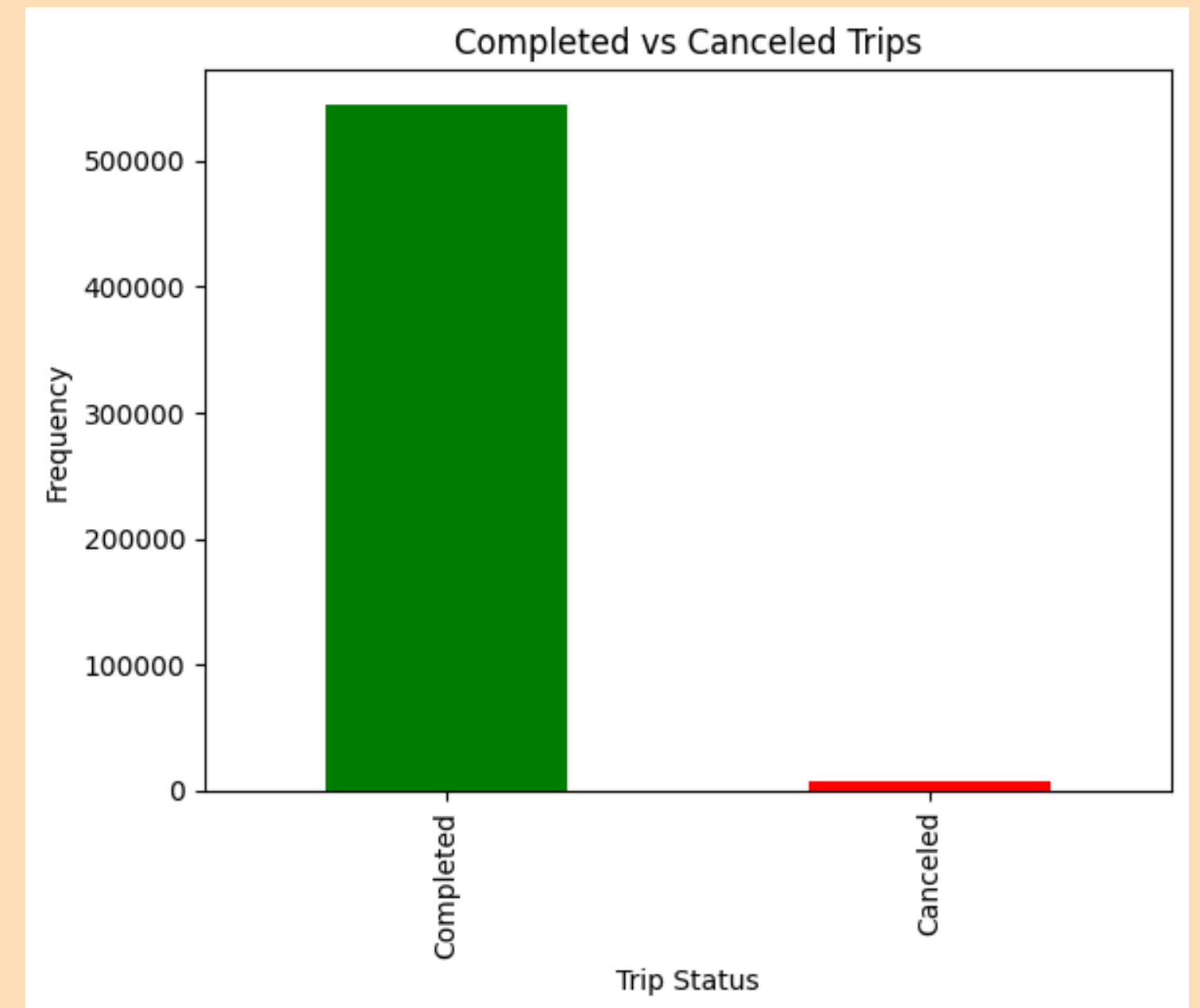
- **Most Trips Are Short:** The majority fall in the 0–5 miles range, with a peak around 1–2 miles, typical for city taxis.
- **Longer Trips Are Rare:** Trips over 5 miles are uncommon, with few exceeding 15 miles.
- **Right-Skewed Distribution:** The histogram shows a long tail, indicating a small number of longer trips.
- **Urban Taxi Behavior:** The data reflects typical urban travel, with short trips dominating and rare long trips like airport transfers.



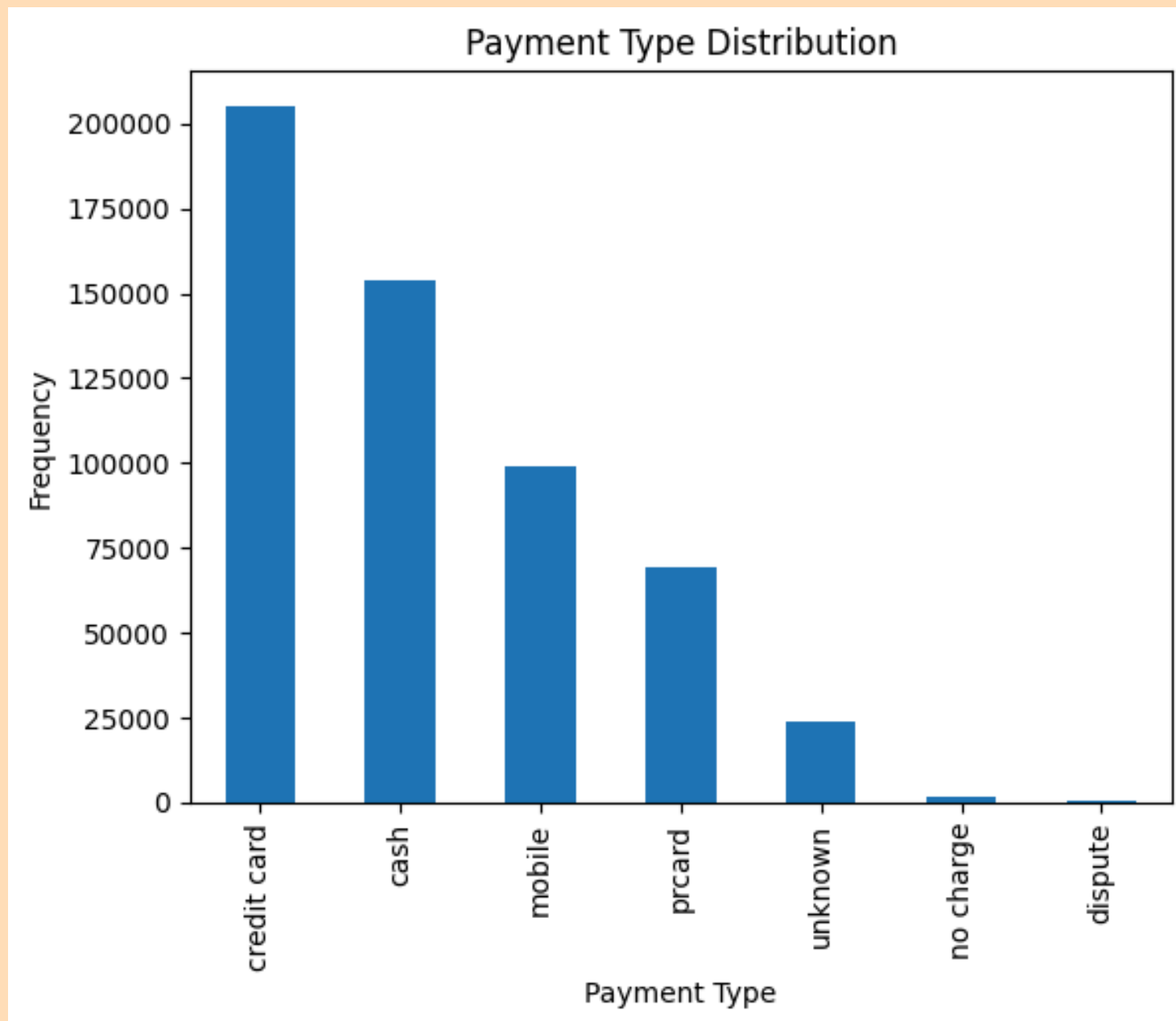
- **Most Trips Are Short:** The majority of trips last 5–20 minutes, peaking around 10 minutes, typical for urban taxi services.
- **Decline in Longer Durations:** Trips longer than 20 minutes become less frequent, with very few exceeding 40 minutes.
- **Right-Skewed Distribution:** The histogram shows a long tail, indicating rare longer trips.



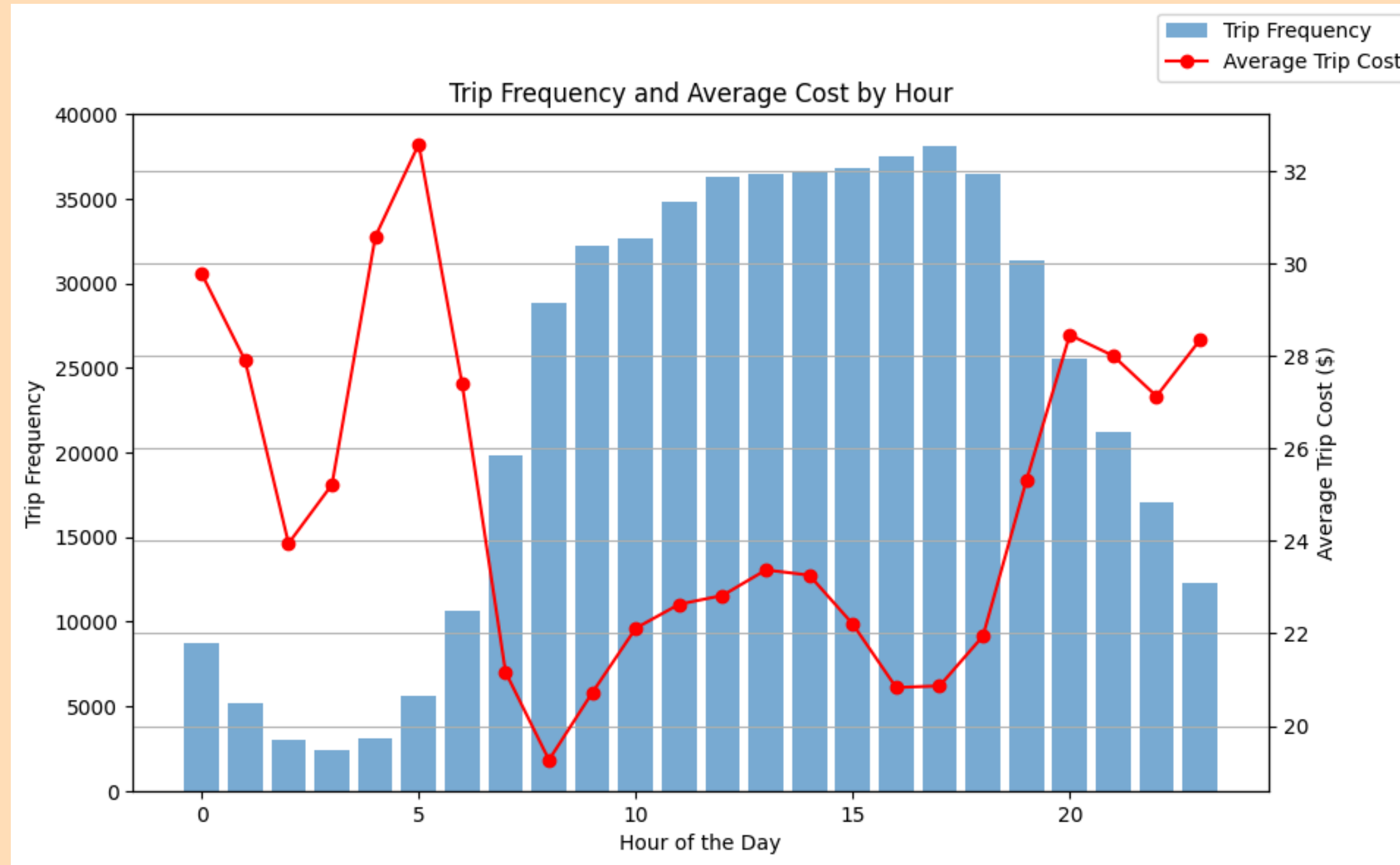
- **Median (Green Line):** The median trip total is around \$15–20, with half of the trips costing less.
- **Interquartile Range (Box):** Most trip totals fall between \$10 and \$32.7, indicating reasonably priced trips.
- **Whiskers:** The whiskers extend from near \$0 to just above \$60, capturing the non-outlier range.
- **Outliers (Black Dots):** A few trips exceed \$60, likely due to long distances, high tips, or surcharges.



- **Most Trips Are Completed:** The green bar shows that most trips are completed, indicating efficient taxi operations and a low cancellation rate.
- **Few Canceled Trips:** The red bar represents a small number of canceled trips, aligning with the expectation that cancellations are rare in urban taxi services.

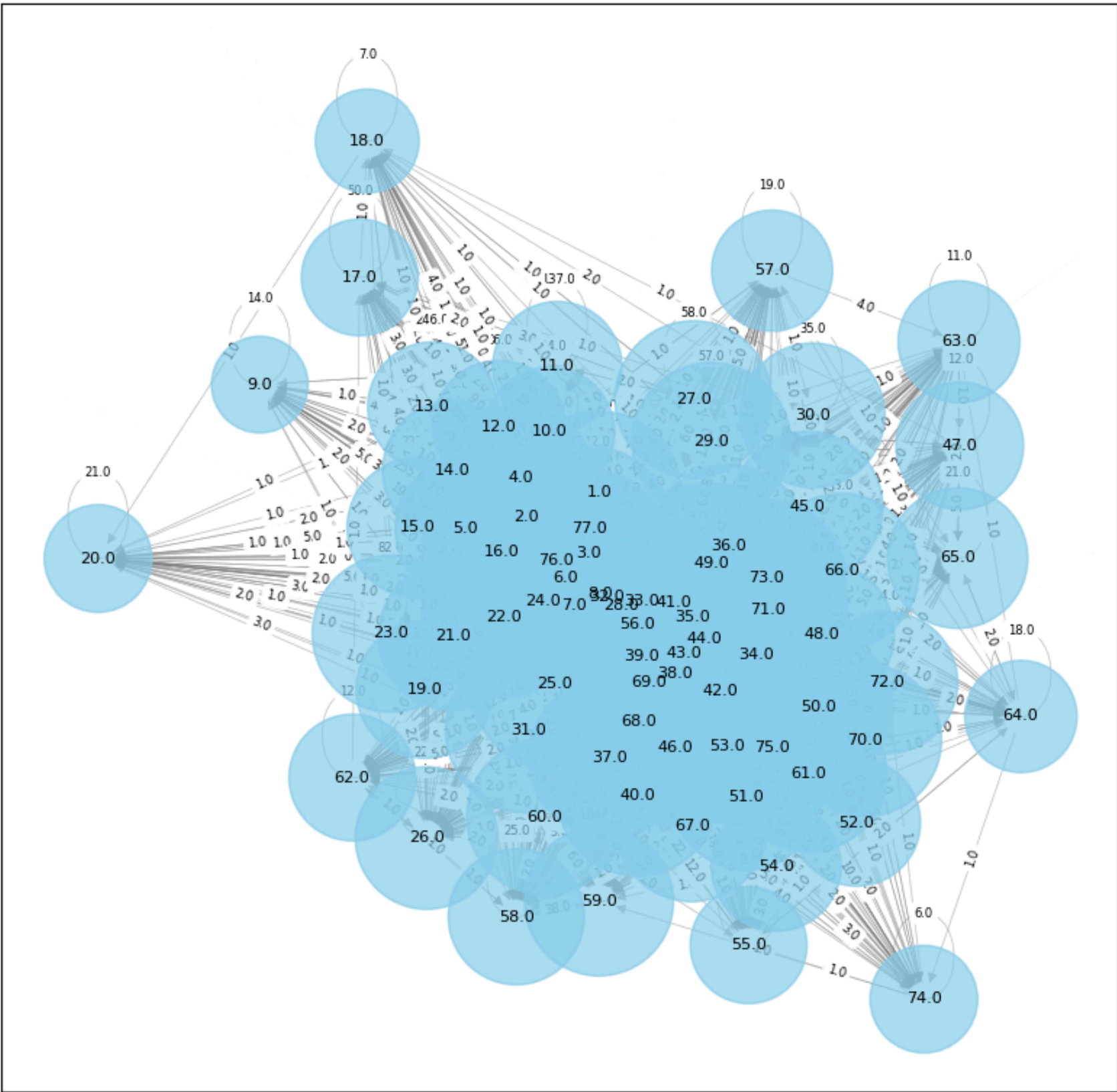


- **Credit Cards Dominate:** Most trips (over 175,000) are paid with credit cards, indicating a preference for electronic payments.
- **Cash as the Second Choice:** Cash is the second most common payment method, showing some passengers still use traditional payment.
- **Mobile Payments and Others:** Mobile payments are growing, while categories like "Prcard," "unknown," and "no charge" are less frequent, possibly due to errors or special cases.
- **Unusual Categories:** Rare categories like "dispute" and "no charge" suggest exceptional cases such as payment issues or promotional rides.

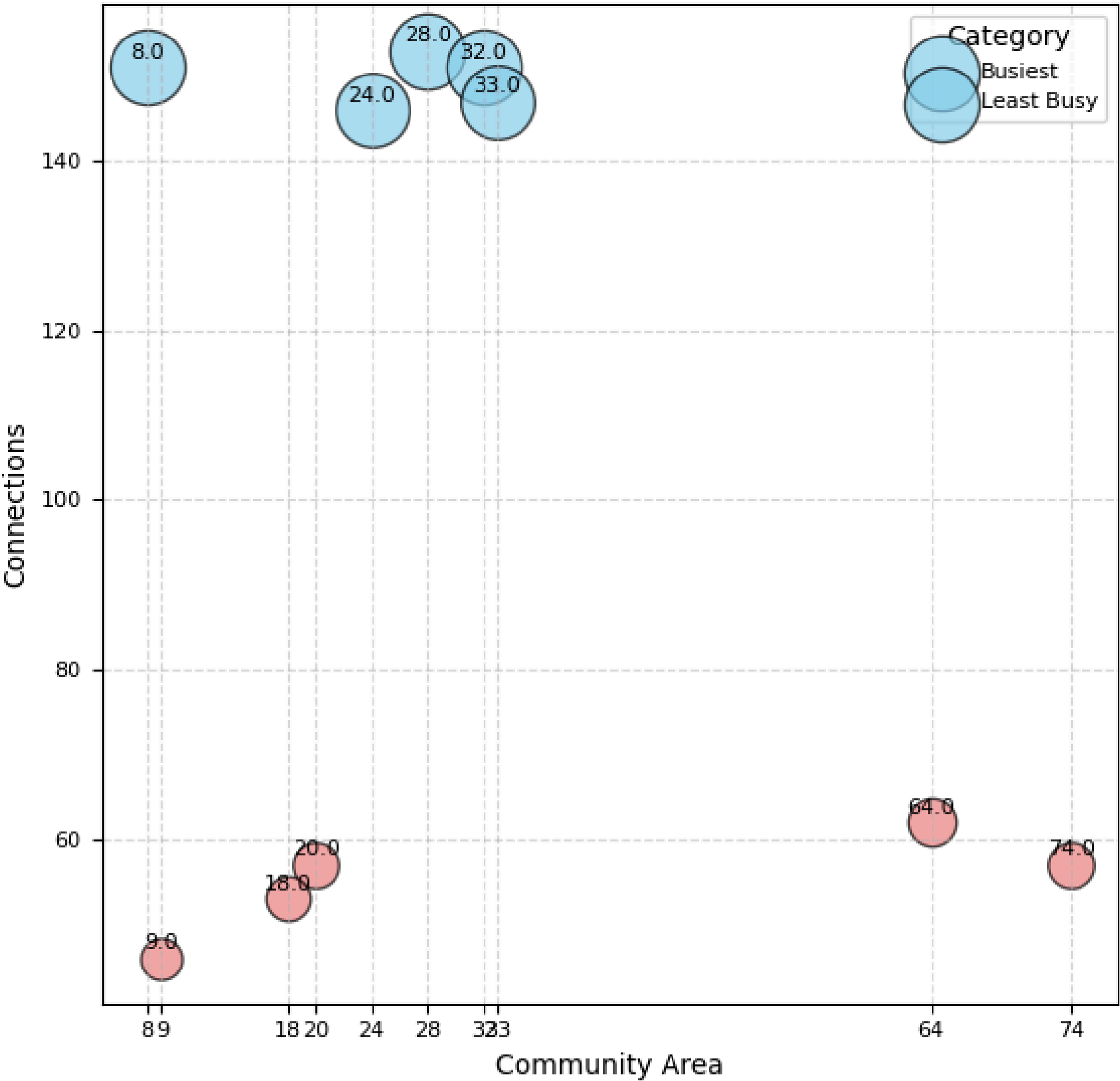


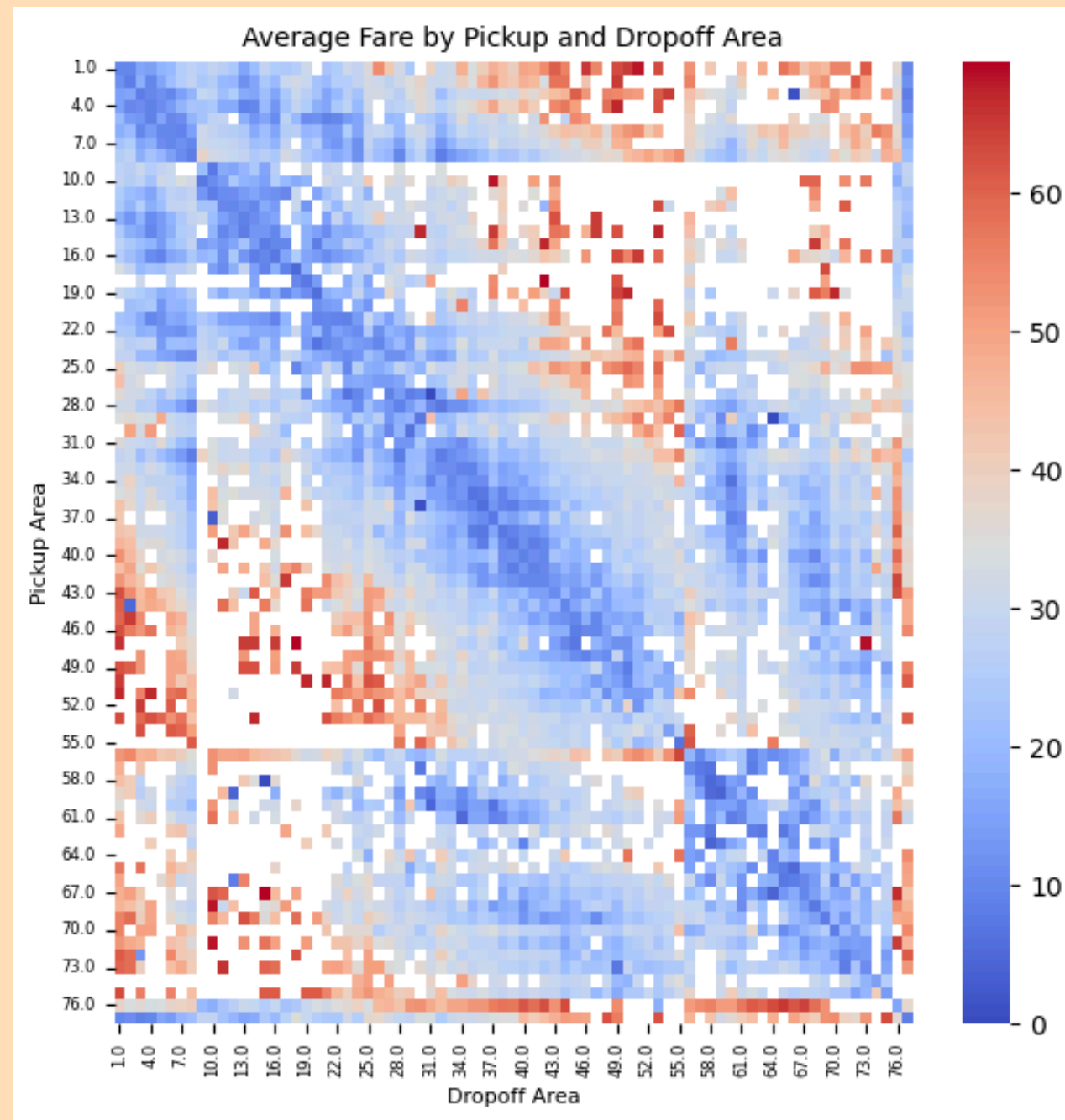
- **Demand Patterns:** Peak demand occurs from 7 AM–10 AM and 4 PM–8 PM, suggesting the need for more taxis during these hours.
- **Revenue Opportunities:** The midnight to early morning period offers higher revenue per trip, indicating long-distance or premium rides.
- **Customer Behavior:** Commuters drive lower average costs during peak hours, while leisure and long-distance travelers increase costs late at night.

Community Area Interaction Network

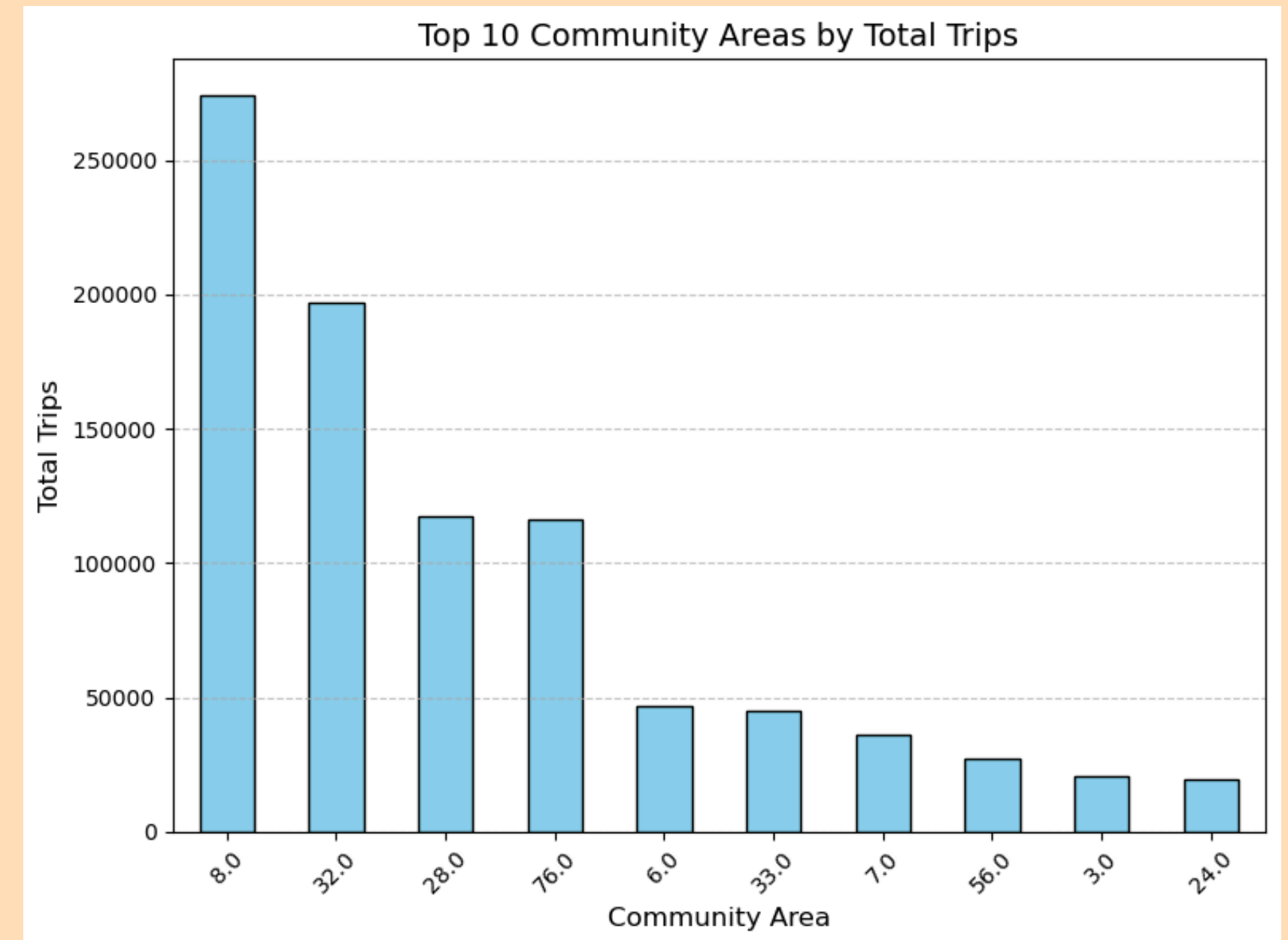


Busiest and Least Busy Community Areas

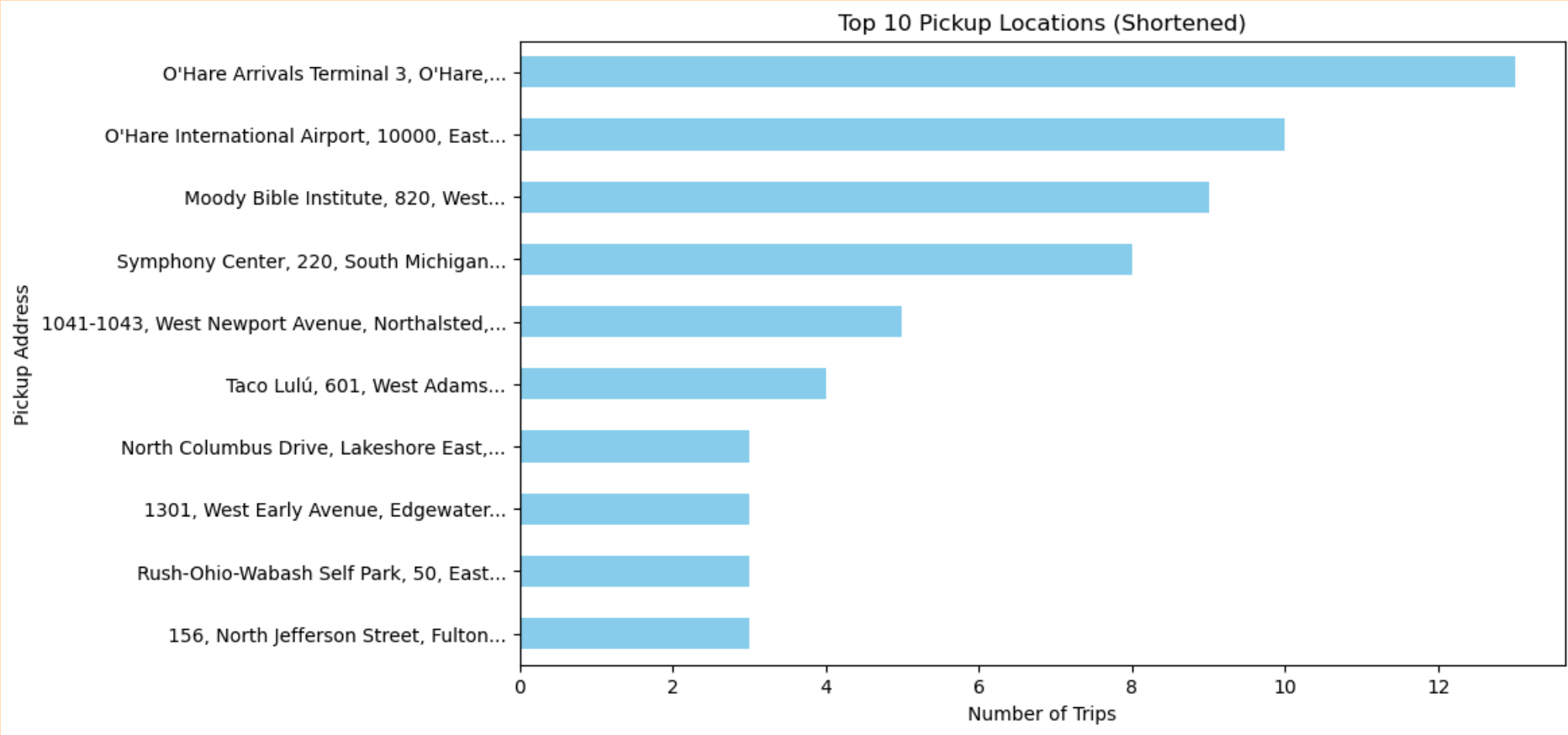




- **High Fare Zones:** Red areas show high fares, likely from long trips or premium locations like airports.
- **Low Fare Zones:** Blue areas represent low fares from short-distance trips.
- **Sparse Data:** White spaces suggest fewer trips between certain areas.



- **High Trip Volume:** Community Area 8.0 leads with the highest number of trips, followed by 32.0 and 76.0, indicating high activity in these regions.
- **Moderate to Low Volume:** Community Areas like 7.0, 56.0, and 24.0 see fewer trips, likely reflecting less demand or smaller populations.

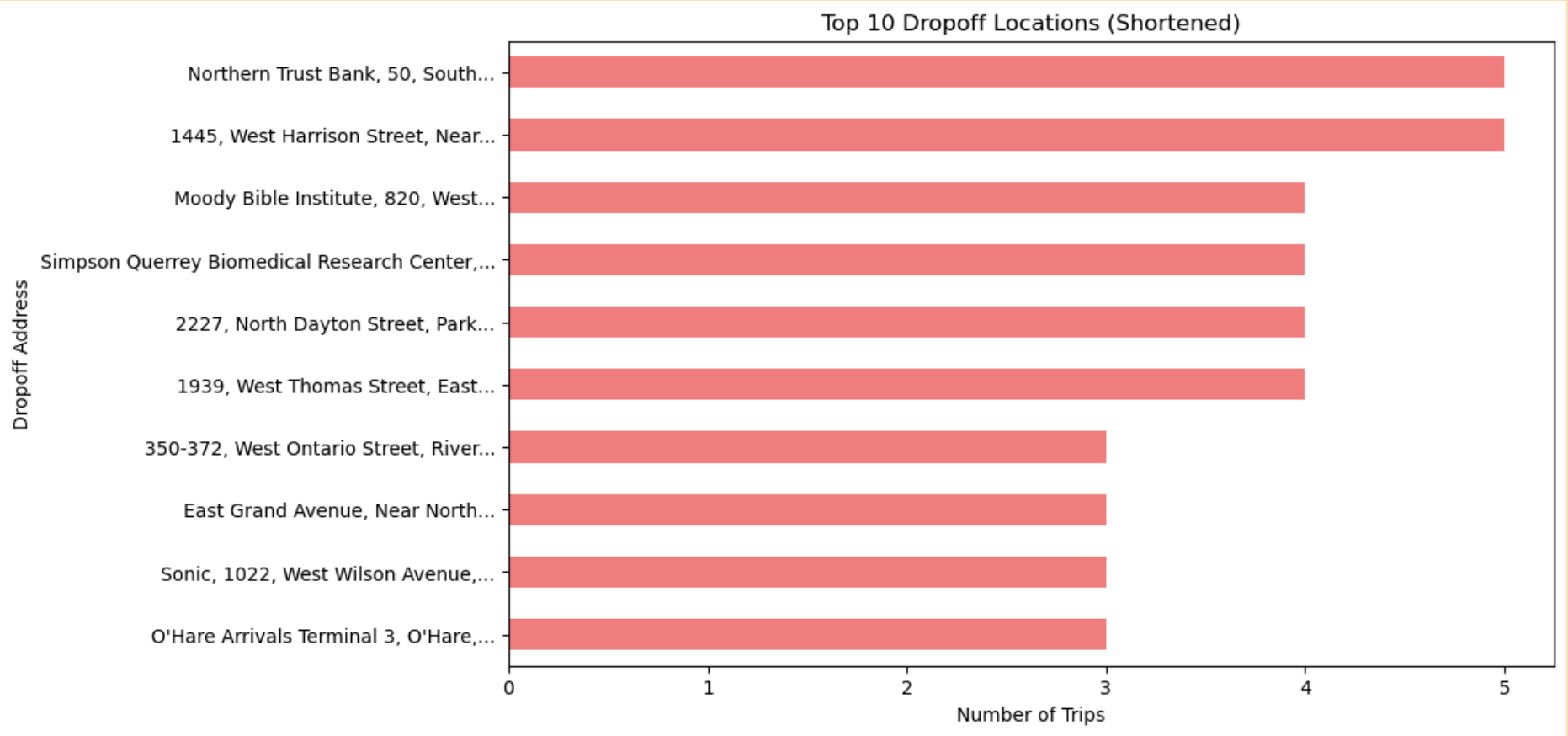


High-Demand Pickup Locations:

- O'Hare Arrivals Terminal 3 and O'Hare International Airport dominate the pickup locations, emphasizing the role of airports as major transportation hubs.
- Other key pickup locations, such as the Moody Bible Institute and Symphony Center, reflect demand from cultural, educational, and entertainment areas.

Geo-Coding Contribution:

- Latitude and longitude data for pickup points were reverse-geocoded to extract readable addresses.
- These addresses were truncated to provide a concise summary while retaining meaningful location identifiers.

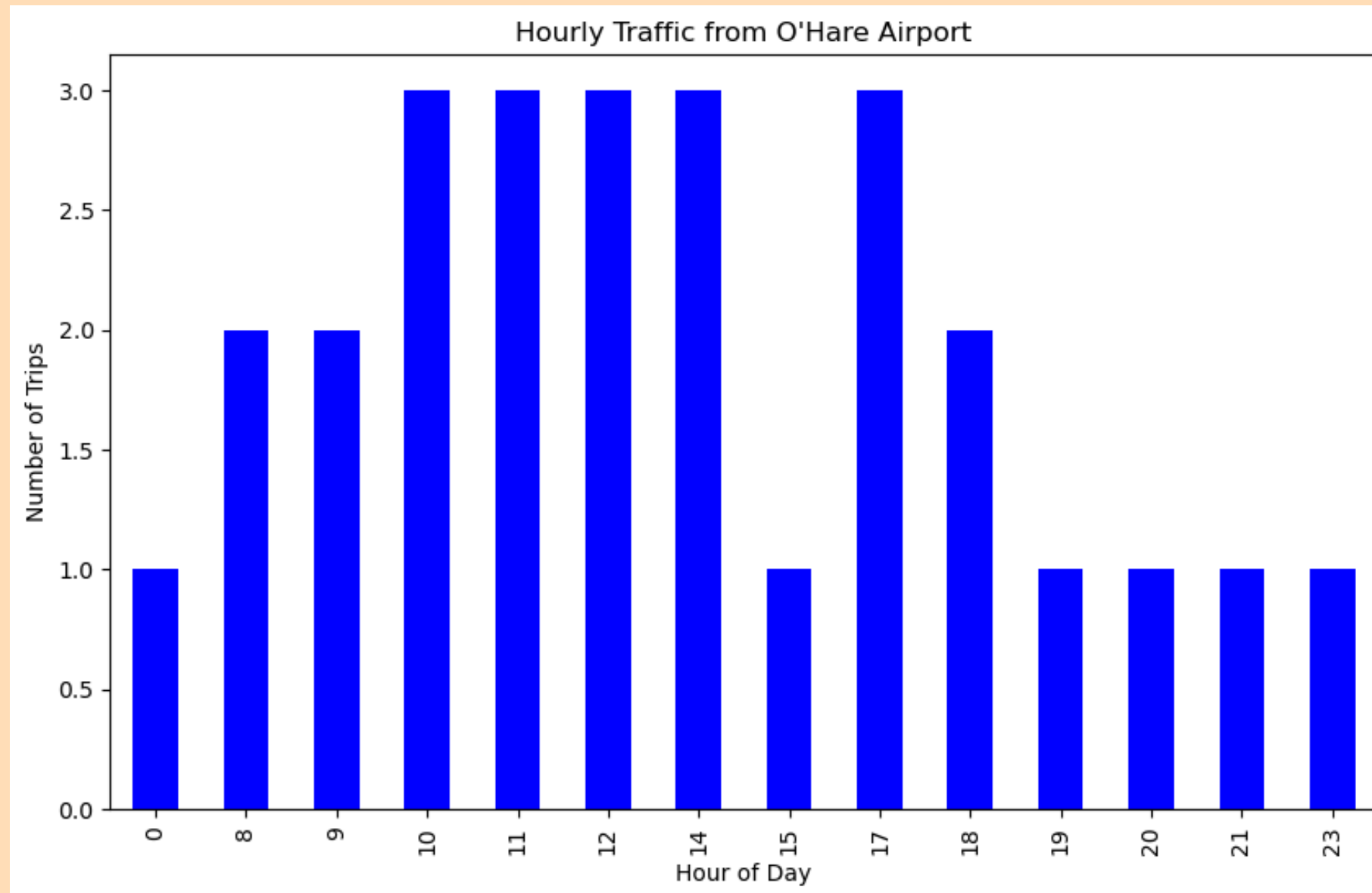


Frequent Dropoff Locations:

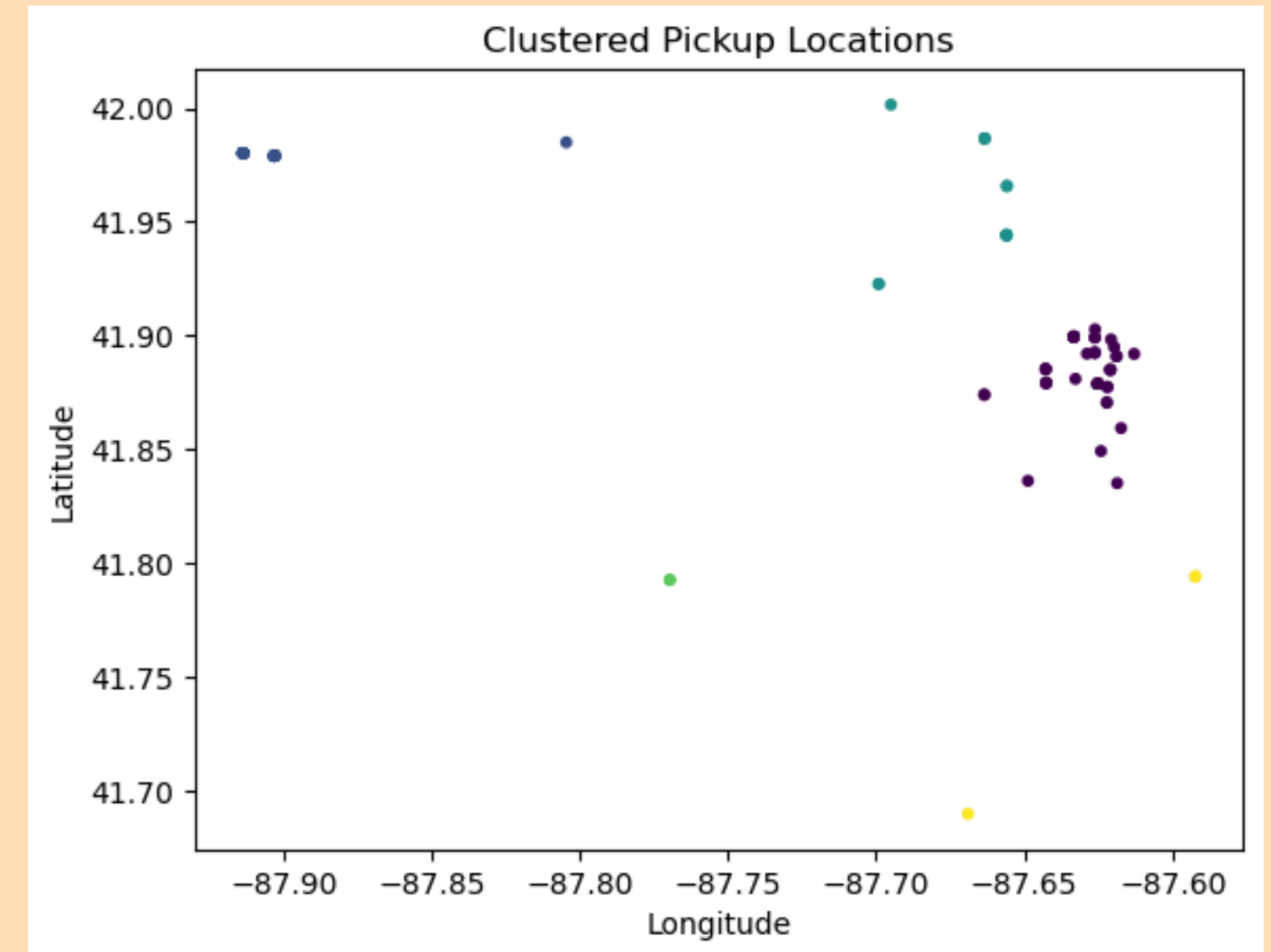
- Northern Trust Bank and 1445 West Harrison Street are prominent dropoff points, likely reflecting areas of economic activity and residential destinations.
- The Moody Bible Institute appears again as a key drop off point, showcasing its centrality in the taxi traffic network.

Geo-Coding Contribution:

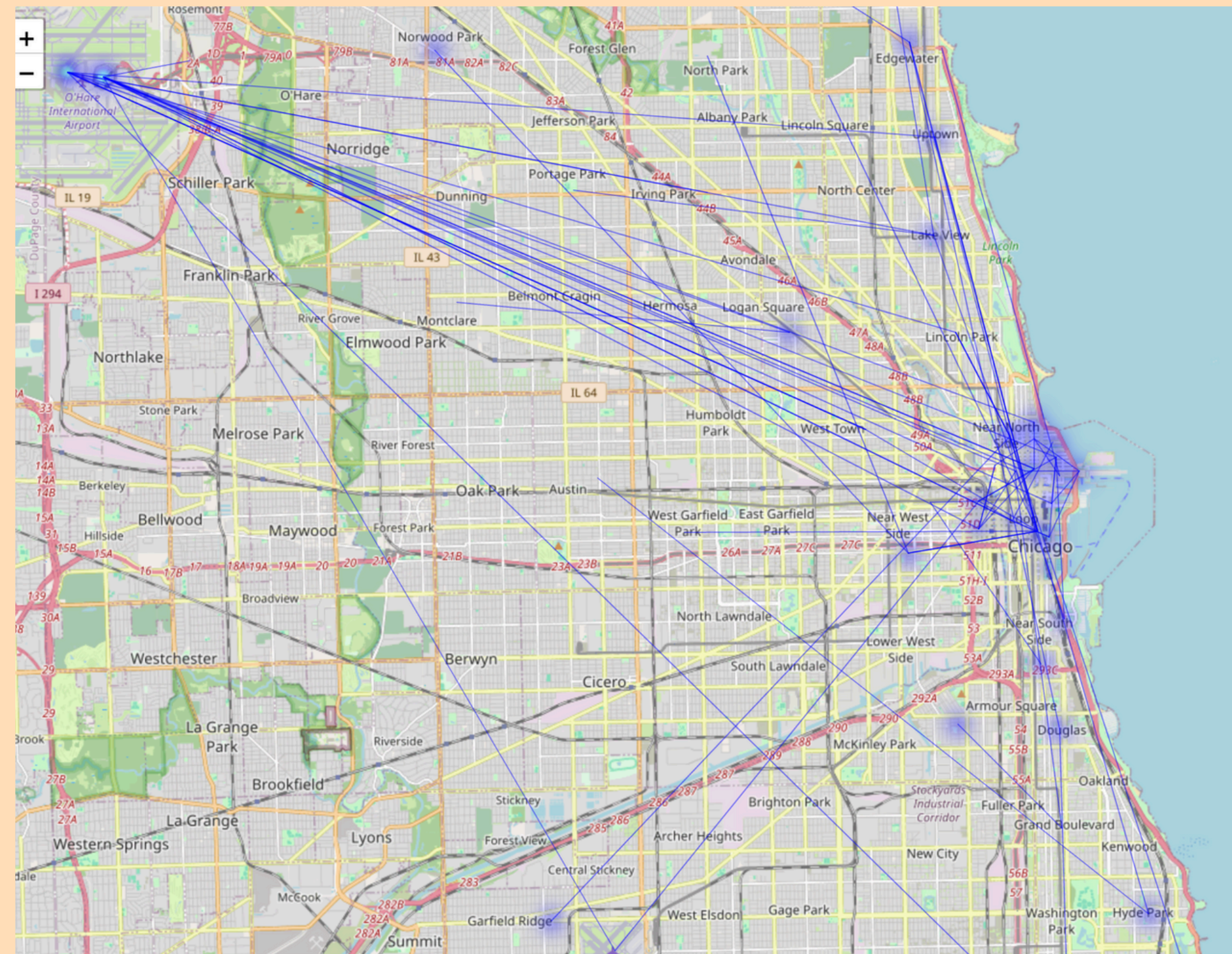
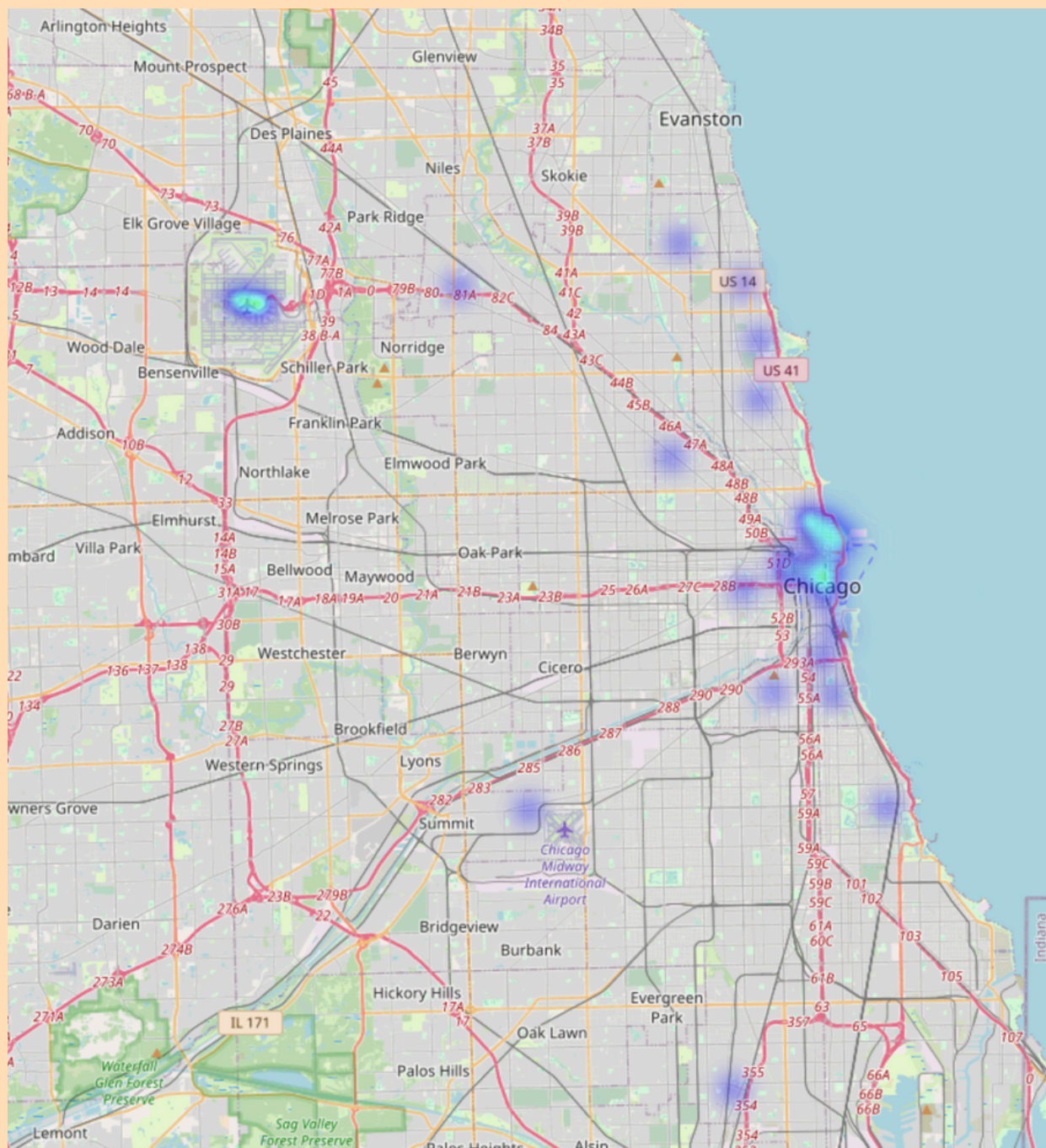
- Dropoff coordinates were reverse-geocoded into human-readable addresses, enabling meaningful interpretation of key destinations.
- Truncated addresses ensure the data remains clear and easy to visualize.



- **Peak Traffic Hours:** The highest taxi traffic occurs between 10 AM and 2 PM, likely due to morning flights and business activity.
- **Evening Spike:** Another noticeable increase in trips is observed around 6 PM, aligning with evening travel demand.
- **Low Traffic Periods:** Traffic drops significantly during the late-night hours, reflecting lower demand.



- **Geographical Concentration:** Pickup locations are densely clustered near central regions (indicated by the concentration of purple points).
- **Outlier Pickups:** A few points, like the yellow markers, represent distant or less frequent pickup locations, possibly serving remote areas or airports.
- **Spatial Patterns:** The clustering suggests high taxi demand in urban centers, while peripheral locations have sporadic pickups.



- **Traffic Corridors:** The flow map highlights major taxi trip corridors, especially from central Chicago and O'Hare Airport, indicating high interconnectivity between these regions.
- **Regional Dynamics:** Thicker lines represent higher traffic volumes, emphasizing significant demand between urban centers and surrounding suburbs.

- **Hotspots:** The heat map reveals concentrated taxi pickups in downtown Chicago and O'Hare Airport, reaffirming their importance as key transportation hubs.
- **Sparse Activity:** Peripheral areas show lower activity, indicating less frequent pickups, likely due to reduced demand or service availability in these regions.

BI QUESTIONS

- **AVERAGE TRIP METRICS: WHAT ARE THE AVERAGE TRIP DISTANCE AND DURATION FOR COMPLETED RIDES?**
- **PEAK REVENUE HOURS: WHICH HOURS GENERATE THE HIGHEST REVENUE AND TRIP FREQUENCY?**
- **PAYMENT METHOD ANALYSIS: WHAT IS THE DISTRIBUTION OF PAYMENT TYPES, AND WHICH GENERATES THE MOST REVENUE?**
- **TRIP CANCELLATIONS: DO CANCELLATIONS OCCUR MORE FREQUENTLY FOR SPECIFIC TRIP LENGTHS OR DURATIONS?**
- **TIP PATTERNS: HOW DO TIP AMOUNTS VARY BASED ON DISTANCE, FARE, AND TIME OF DAY?**

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THANK YOU
ANY QUESTIONS?